

# Linear Algebra

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## Vectors

**Exercise 0.1** (Store a data point as one object). A house has a floor area of 120 square meters, 3 bedrooms, and an age of 15 years.

1. Write these three numbers as a single vertical list  $\tilde{x}$ , in the order given, and call its entries  $x_1$ ,  $x_2$  and  $x_3$ . What is  $x_2$ ?
2. Does the list  $(3, 120, 15)$  describe the same house?

### Solution

*Solution 0.1.*

1. Stacking the three numbers in order gives

$$\tilde{x} = \begin{bmatrix} 120 \\ 3 \\ 15 \end{bmatrix},$$

so  $x_2 = 3$ , the number of bedrooms.

2. No. The list  $(3, 120, 15)$  would describe a house with 3 square meters of floor area and 120 bedrooms. An entry's position says what it measures, so the order of the entries matters.

**Definition 0.1** (Column vector). A **column vector** of length  $p$  is an ordered list of  $p$  numbers, written vertically:

$$\tilde{x} = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_p \end{bmatrix}$$

*Remark 0.1* (Vectors are columns by default). In these notes, as in most statistics textbooks, a vector is a column vector unless stated otherwise. A vector written inline as a list in parentheses is still a column vector: for example,  $\tilde{x} = (2, -1, 5)$  means

$$\tilde{x} = \begin{bmatrix} 2 \\ -1 \\ 5 \end{bmatrix}.$$

**Definition 0.2** (The set  $\mathbb{R}^p$ ). For a positive integer  $p$ ,  $\mathbb{R}^p$  is the set of all **column vectors** of length  $p$  whose entries are real numbers.

*Remark 0.2* (Reading  $\tilde{x} \in \mathbb{R}^p$ ). Writing  $\tilde{x} \in \mathbb{R}^p$  says, in one symbol, that  $\tilde{x}$  is a column vector with  $p$  real entries. For example,  $(1, -2, 0.5) \in \mathbb{R}^3$ . The vector  $(1, 2)$  is not in  $\mathbb{R}^3$ , because it has only 2 entries; it is in  $\mathbb{R}^2$ . In Exercise 0.1,  $\tilde{x} \in \mathbb{R}^3$ .

**Definition 0.3** (Transpose). The **transpose** of a column vector  $\tilde{x}$  is the row vector with the same sequence of entries, written horizontally:

$$\tilde{x}^\top \equiv \tilde{x}' \equiv [x_1, x_2, \dots, x_p]$$

**Example 0.1** (Transposing a column vector). The transpose of the column vector with entries 2, -1, 5 is the row vector with the same entries:

$$\begin{bmatrix} 2 \\ -1 \\ 5 \end{bmatrix}^\top = [2, -1, 5].$$

**Definition 0.4** (Vector addition). The **sum** of two column vectors  $\tilde{x}$  and  $\tilde{y}$  of the same length  $p$  is the column vector  $\tilde{x} + \tilde{y}$  of length  $p$  obtained by adding entry by entry:

$$(\tilde{x} + \tilde{y})_i \stackrel{\text{def}}{=} x_i + y_i, \quad i = 1, \dots, p$$

**Example 0.2** (Adding two vectors).

$$\begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} + \begin{bmatrix} 4 \\ 5 \\ 6 \end{bmatrix} = \begin{bmatrix} 1+4 \\ 2+5 \\ 3+6 \end{bmatrix} = \begin{bmatrix} 5 \\ 7 \\ 9 \end{bmatrix}$$

**Definition 0.5** (Dot product). For any two real-valued vectors  $\tilde{x} = (x_1, \dots, x_p)$  and  $\tilde{y} = (y_1, \dots, y_p)$  of the same length  $p$ , the **dot product** of  $\tilde{x}$  and  $\tilde{y}$  is:

$$\tilde{x} \cdot \tilde{y} = \tilde{x}^\top \tilde{y} \stackrel{\text{def}}{=} \sum_{i=1}^p x_i y_i \quad (1)$$

See also the definitions in:

- Dobson and Barnett (2018), Section 1.3 (equation 1.1, page 7)
- Kaplan (2022), chapter on vectors
- [wikipedia](#)

The dot product has a different generalization for two matrices; see [wikipedia](#) for more.

**Example 0.3** (A dot product). For  $\tilde{x} = (1, 2, 3)$  and  $\tilde{y} = (4, 5, 6)$ :

$$\begin{aligned} \tilde{x} \cdot \tilde{y} &= 1 \cdot 4 + 2 \cdot 5 + 3 \cdot 6 && \text{(definition of the dot product)} \\ &= 4 + 10 + 18 && \text{(multiply)} \\ &= 32 && \text{(add)} \end{aligned}$$

**Definition 0.6** (Linear combination). A **linear combination** of the numbers  $a_1, \dots, a_k$  with **coefficients**  $c_1, \dots, c_k$  is the weighted sum

$$c_1 a_1 + \dots + c_k a_k = \sum_{i=1}^k c_i a_i$$

A linear combination of vectors  $\tilde{v}_1, \dots, \tilde{v}_k$  of the same length is defined entry by entry: entry  $j$  of  $\sum_{i=1}^k c_i \tilde{v}_i$  is the linear combination of the  $j$ th entries of  $\tilde{v}_1, \dots, \tilde{v}_k$  with the same coefficients. For example, the linear combination of  $(1, 0)^\top$  and  $(0, 1)^\top$  with coefficients 2 and 3 is  $(2 \cdot 1 + 3 \cdot 0, 2 \cdot 0 + 3 \cdot 1)^\top = (2, 3)^\top$ .

*Remark 0.3* (The dot product is a linear combination). The dot product  $\tilde{x} \cdot \tilde{y}$  is a linear combination (Definition 0.6) of the entries of  $\tilde{y}$ , with coefficients  $x_1, \dots, x_p$ . In Example 0.3,

$\tilde{x} \cdot \tilde{y} = 1 \cdot 4 + 2 \cdot 5 + 3 \cdot 6$  is the linear combination of 4, 5, and 6 with coefficients 1, 2, and 3.

The dot product is also the standard *inner product* on  $\mathbb{R}^p$ ; “inner product” is the general notion, of which the dot product is one example.

**Theorem 0.1** (Dot product is symmetric). *The dot product is symmetric:*

$$\tilde{x} \cdot \tilde{y} = \tilde{y} \cdot \tilde{x}$$

**i** Proof

*Proof.*

$$\begin{aligned} \tilde{x} \cdot \tilde{y} &= \sum_{i=1}^p x_i y_i && \text{(definition of the dot product)} \\ &= \sum_{i=1}^p y_i x_i && \text{(products of numbers are symmetric)} \\ &= \tilde{y} \cdot \tilde{x} && \text{(definition of the dot product)} \end{aligned}$$

□

## Special vectors

**Definition 0.7** (Zero vector). The **zero vector**  $\tilde{0}$  of length  $p$  has all entries equal to zero:

$$\tilde{0} = \begin{bmatrix} 0 \\ 0 \\ \vdots \\ 0 \end{bmatrix}$$

*Remark 0.4* (The zero vector is the additive identity). Adding the zero vector to a vector leaves it unchanged (Definition 0.4):  $\tilde{x} + \tilde{0} = \tilde{x}$  for any vector  $\tilde{x}$  of the same length. For example,  $(2, -1) + (0, 0) = (2 + 0, -1 + 0) = (2, -1)$ .

**Definition 0.8** (Ones vector). The **ones vector**  $\tilde{1}$  of length  $p$  has all entries equal to one:

$$\tilde{\mathbf{1}} = \begin{bmatrix} 1 \\ 1 \\ \vdots \\ 1 \end{bmatrix}$$

**Remark 0.5** (The ones vector sums the entries). The dot product (Definition 0.5) of  $\tilde{\mathbf{1}}$  with  $\tilde{x}$  is the sum of the entries of  $\tilde{x}$ :  $\tilde{\mathbf{1}} \cdot \tilde{x} = \sum_{i=1}^p 1 \cdot x_i = \sum_{i=1}^p x_i$ . For example, for  $\tilde{x} = (2, -1, 5)$ ,  $\tilde{\mathbf{1}} \cdot \tilde{x} = 2 + (-1) + 5 = 6$ , and dividing by the length  $p = 3$  gives the mean of the entries,  $6/3 = 2$ .

**Definition 0.9** (Indicator vector / standard basis vector). The  $j$ -th **indicator vector** (or *standard basis vector*)  $\tilde{e}_j$  of length  $p$  has a 1 in position  $j$  and 0s elsewhere:

$$(\tilde{e}_j)_i = \begin{cases} 1 & \text{if } i = j \\ 0 & \text{if } i \neq j \end{cases} \quad \tilde{e}_j = \begin{bmatrix} 0 \\ \vdots \\ 0 \\ 1 \\ 0 \\ \vdots \\ 0 \end{bmatrix} \leftarrow \text{position } j$$

**Remark 0.6** (Indicator vectors build every vector). For  $p = 3$ , the indicator vectors are  $\tilde{e}_1 = (1, 0, 0)$ ,  $\tilde{e}_2 = (0, 1, 0)$ , and  $\tilde{e}_3 = (0, 0, 1)$ . Every vector of length 3 is a weighted sum  $c_1\tilde{e}_1 + c_2\tilde{e}_2 + c_3\tilde{e}_3$  of them, with weights equal to its entries; for example,  $(2, -1, 5) = 2\tilde{e}_1 - 1\tilde{e}_2 + 5\tilde{e}_3$ . No other weights work: entry  $j$  of  $c_1\tilde{e}_1 + c_2\tilde{e}_2 + c_3\tilde{e}_3$  is  $c_j$ , so each weight  $c_j$  must equal entry  $j$  of the vector.

**Theorem 0.2** (Indicator vectors select entries). *For any vector  $\tilde{x}$  of length  $p$  and any  $j \in \{1, \dots, p\}$ :*

$$\tilde{e}_j \cdot \tilde{x} = x_j$$

### **i** Proof

*Proof.* Writing the dot product componentwise:

$$\begin{aligned}
\tilde{e}_j \cdot \tilde{x} &= \sum_{i=1}^p (\tilde{e}_j)_i x_i && \text{(definition of the dot product)} \\
&= \sum_{i=1}^p \begin{cases} 1 \cdot x_i & \text{if } i = j \\ 0 \cdot x_i & \text{if } i \neq j \end{cases} && \text{(definition of } \tilde{e}_j \text{)} \\
&= x_j && \text{(only the } i = j \text{ term is nonzero)}
\end{aligned}$$

□

## Orthogonality

**Definition 0.10** (Orthogonal vectors). Two vectors  $\tilde{x}$  and  $\tilde{y}$  of the same length are **orthogonal** (written  $\tilde{x} \perp \tilde{y}$ ) if their dot product (Definition 0.5) is zero:

$$\tilde{x} \perp \tilde{y} \iff \tilde{x} \cdot \tilde{y} = 0$$

*Remark 0.7* (Orthogonal means perpendicular). Orthogonality extends the geometric idea of perpendicular lines to vectors with any number of entries. In the plane,  $(1, 2)$  and  $(-2, 1)$  are perpendicular, and their dot product is  $1 \cdot (-2) + 2 \cdot 1 = 0$ . The same test works in three dimensions, where pictures are harder to draw:  $(1, 1, 0) \cdot (1, -1, 5) = 1 - 1 + 0 = 0$ , so  $(1, 1, 0) \perp (1, -1, 5)$ .

**Definition 0.11** (Euclidean norm). The **Euclidean norm** (or *length*, also called the  $L_2$  norm and written  $\|\tilde{x}\|_2$ ) of a vector  $\tilde{x}$  of length  $p$  is

$$\|\tilde{x}\| \stackrel{\text{def}}{=} \sqrt{\tilde{x} \cdot \tilde{x}} = \sqrt{\sum_{i=1}^p x_i^2} \quad (2)$$

**Example 0.4** (The length of a vector). For  $\tilde{x} = (3, 4)$ :

$$\begin{aligned}
\|\tilde{x}\| &= \sqrt{3^2 + 4^2} && \text{(definition of the norm)} \\
&= \sqrt{25} && \text{(square and add)} \\
&= 5
\end{aligned}$$

The vector  $(0.6, 0.8)$  has norm  $\sqrt{0.36 + 0.64} = 1$ .

**Exercise 0.2** (How far apart are two points?). Let  $\tilde{x} = (1, 2)$  and  $\tilde{y} = (4, 6)$ .

1. Compute the vector  $\tilde{x} - \tilde{y}$ .
2. Compute the length of  $\tilde{x} - \tilde{y}$ , using the [Euclidean norm](#).
3. Is the result the same if you compute the length of  $\tilde{y} - \tilde{x}$  instead?

Solution

*Solution 0.2.*

1.  $\tilde{x} - \tilde{y} = (1 - 4, 2 - 6) = (-3, -4)$ .
2.  $\|\tilde{x} - \tilde{y}\| = \sqrt{(-3)^2 + (-4)^2} = \sqrt{25} = 5$ .
3.  $\tilde{y} - \tilde{x} = (3, 4)$ , and  $\|\tilde{y} - \tilde{x}\| = \sqrt{3^2 + 4^2} = 5$ . Yes, the result is the same.

**Definition 0.12** (Euclidean distance). The **Euclidean distance** between two vectors  $\tilde{x}$  and  $\tilde{y}$  of length  $p$  is the [Euclidean norm](#) of their difference:

$$d(\tilde{x}, \tilde{y}) \stackrel{\text{def}}{=} \|\tilde{x} - \tilde{y}\| = \sqrt{\sum_{i=1}^p (x_i - y_i)^2} \quad (3)$$

*Remark 0.8* (Distance from the origin). The distance from the origin  $\tilde{0}$  to a vector  $\tilde{x}$  is the norm of  $\tilde{x}$ :  $d(\tilde{x}, \tilde{0}) = \|\tilde{x} - \tilde{0}\| = \|\tilde{x}\|$  (Goodfellow et al. 2016, 39).

For example, the distance from  $\tilde{0}$  to  $(3, 4)$  is  $\|(3, 4)\| = 5$ , as in [Example 0.4](#).

**Definition 0.13** (Orthonormal vectors). A set of vectors  $\{\tilde{x}_1, \tilde{x}_2, \dots, \tilde{x}_k\}$  is **orthonormal** if the vectors are mutually orthogonal ([Definition 0.10](#)) and each has norm 1 ([Definition 0.11](#)), that is, unit length:

$$\tilde{x}_i \cdot \tilde{x}_j = \begin{cases} 1 & \text{if } i = j \\ 0 & \text{if } i \neq j \end{cases}$$

**Example 0.5** (Indicator vectors are orthonormal). The indicator vectors  $\tilde{e}_1, \tilde{e}_2, \dots, \tilde{e}_p$  ([Definition 0.9](#)) form an orthonormal set. By [Theorem 0.2](#),  $\tilde{e}_i \cdot \tilde{e}_j$  is entry  $i$  of  $\tilde{e}_j$ , which is 1 if  $i = j$  and 0 if  $i \neq j$ . For  $p = 2$ :  $\tilde{e}_1 \cdot \tilde{e}_1 = 1 \cdot 1 + 0 \cdot 0 = 1$ ,  $\tilde{e}_2 \cdot \tilde{e}_2 = 0 \cdot 0 + 1 \cdot 1 = 1$ , and  $\tilde{e}_1 \cdot \tilde{e}_2 = 1 \cdot 0 + 0 \cdot 1 = 0$ .

**Exercise 0.3** (Compute an inner product and a norm). Let

$$\tilde{a} = \begin{bmatrix} 3 \\ -1 \\ 2 \end{bmatrix}, \quad \tilde{b} = \begin{bmatrix} 0 \\ 4 \\ -2 \end{bmatrix}$$

Compute  $\tilde{a}^\top \tilde{b}$  and  $\|\tilde{a}\|_2$ .

Solution

*Solution 0.3.* Multiply entry by entry and add:

$$\tilde{a}^\top \tilde{b} = (3)(0) + (-1)(4) + (2)(-2) = 0 - 4 - 4 = -8$$

For the norm, take the inner product of  $\tilde{a}$  with itself first:

$$\tilde{a}^\top \tilde{a} = 3^2 + (-1)^2 + 2^2 = 9 + 1 + 4 = 14$$

so

$$\|\tilde{a}\|_2 = \sqrt{14} \approx 3.742$$

The inner product came out negative while the norm cannot: a norm is a square root of a sum of squares, so it is never below zero.

Python has both operations built in, and R builds them from `sum`.

```
import numpy as np

a = np.array([3.0, -1.0, 2.0])
b = np.array([0.0, 4.0, -2.0])

print(a @ b, round(float(np.linalg.norm(a)), 3))
```

```
a <- c(3, -1, 2)
b <- c(0, 4, -2)

print(c(sum(a * b), round(sqrt(sum(a^2)), 3)))
#> [1] -8.000 3.742
```

Both print  $-8$  and  $3.742$ .

💡 Video lecture

Hutchinson's [Linear Function Basics](#) (24 min) covers dot products and the Euclidean norm, and goes on to hyperplanes and level sets (Hutchinson, n.d.). The login for the video site is posted [on Canvas](#).

## Matrices

**Definition 0.14** (Matrix). A **matrix** of dimensions  $m \times n$  is a rectangular array of  $m \cdot n$  numbers, arranged in  $m$  rows and  $n$  columns:

$$\mathbf{A} = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix}$$

*Remark 0.9* (Matrix entries, and vectors as matrices). The entry in row  $i$  and column  $j$  of  $\mathbf{A}$  is written  $a_{ij}$  or  $(\mathbf{A})_{ij}$ . For example, for the  $2 \times 3$  matrix

$$\mathbf{A} = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix},$$

$a_{12} = 2$  and  $a_{23} = 6$ .

A column vector of length  $p$  is a  $p \times 1$  matrix, and a row vector of length  $p$  is a  $1 \times p$  matrix. For example, the column vector  $(7, 8)$  is a  $2 \times 1$  matrix, and the row vector  $[7, 8]$  is a  $1 \times 2$  matrix.

**Definition 0.15** (Tensor). A **tensor** of order  $k \geq 1$  is an array of numbers with  $k$  indices,

$$\mathcal{A} = (a_{i_1 i_2 \dots i_k}), \quad i_1 = 1, \dots, n_1, \dots, i_k = 1, \dots, n_k,$$

and its dimensions are  $n_1 \times n_2 \times \cdots \times n_k$ .

**Example 0.6** (Vectors, matrices and images as tensors). A vector of length  $p$  is a tensor of order 1, and an  $m \times n$  [matrix](#) is a tensor of order 2. A  $28 \times 28$  grayscale image is a matrix: the entry  $a_{ij}$  is the brightness of the pixel in row  $i$  and column  $j$ . A  $28 \times 28$  color image stores three numbers per pixel (red, green, blue), so it is an order-3 tensor with dimensions  $28 \times 28 \times 3$ , and  $a_{ijc}$  is the value of color channel  $c$  at the pixel in row  $i$  and column  $j$ . It has  $28 \cdot 28 \cdot 3 = 2352$  entries.

## Matrix transpose

**Definition 0.16** (Matrix transpose). The **transpose** of an  $m \times n$  matrix  $\mathbf{A}$  is the  $n \times m$  matrix  $\mathbf{A}^\top$  obtained by swapping the rows and columns of  $\mathbf{A}$ :

$$(\mathbf{A}^\top)_{ij} = a_{ji}$$

## Matrix addition

**Definition 0.17** (Zero matrix). The  $m \times n$  **zero matrix**  $\mathbf{0}_{m \times n}$  (or  $\mathbf{0}$  when dimensions are clear from context) has all entries equal to zero:

$$\mathbf{0}_{m \times n} = \begin{bmatrix} 0 & 0 & \cdots & 0 \\ 0 & 0 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 0 \end{bmatrix}$$

**Definition 0.18** (Matrix addition). Two matrices  $\mathbf{A}$  and  $\mathbf{B}$  of the same dimensions  $m \times n$  can be added element-wise; their **matrix sum** is:

$$(\mathbf{A} + \mathbf{B})_{ij} = a_{ij} + b_{ij}$$

**Theorem 0.3** (Matrix addition is commutative). For  $m \times n$  matrices  $\mathbf{A}$  and  $\mathbf{B}$ :

$$\mathbf{A} + \mathbf{B} = \mathbf{B} + \mathbf{A}$$

**Theorem 0.4** (Matrix addition is associative). For  $m \times n$  matrices  $\mathbf{A}$ ,  $\mathbf{B}$ , and  $\mathbf{C}$ :

$$(\mathbf{A} + \mathbf{B}) + \mathbf{C} = \mathbf{A} + (\mathbf{B} + \mathbf{C})$$

**Theorem 0.5** (Zero matrix is the additive identity). For an  $m \times n$  matrix  $\mathbf{A}$ :

$$\mathbf{A} + \mathbf{0}_{m \times n} = \mathbf{A}$$

**Theorem 0.6** (Additive inverse). For any  $m \times n$  matrix  $\mathbf{A}$ , the  $m \times n$  matrix  $-\mathbf{A}$  (defined by  $(-\mathbf{A})_{ij} = -a_{ij}$ ) satisfies:

$$\mathbf{A} + (-\mathbf{A}) = \mathbf{0}_{m \times n}$$

## Scalar multiplication

**Definition 0.19** (Scalar multiplication). The **scalar multiple** of a matrix  $\mathbf{A}$  by a scalar  $c$  is:

$$(c\mathbf{A})_{ij} = c \cdot a_{ij}$$

## Matrix multiplication

**Definition 0.20** (Matrix multiplication). The **product** of an  $m \times k$  matrix  $\mathbf{A}$  and a  $k \times n$  matrix  $\mathbf{B}$  is the  $m \times n$  matrix  $\mathbf{C} = \mathbf{AB}$  with entries:

$$c_{ij} = \sum_{s=1}^k a_{is} b_{sj}$$

*Remark 0.10* (Matrix multiplication is not commutative). When  $\mathbf{AB}$  and  $\mathbf{BA}$  are both defined, they are usually different. For example, with

$$\mathbf{A} = \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix}, \quad \mathbf{B} = \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix},$$

$$\mathbf{AB} = \begin{bmatrix} 3 & 2 \\ 1 & 1 \end{bmatrix} \neq \begin{bmatrix} 1 & 2 \\ 1 & 3 \end{bmatrix} = \mathbf{BA}.$$

**Example 0.7** (Dot product as matrix multiplication). The dot product of two column vectors  $\tilde{x}$  and  $\tilde{\beta}$  can be written as a matrix product of the row vector  $\tilde{x}^\top$  with the column vector  $\tilde{\beta}$ :

$$\begin{aligned}
\tilde{x} \cdot \tilde{\beta} &= \tilde{x}^\top \tilde{\beta} \\
&= [x_1, x_2, \dots, x_p] \begin{bmatrix} \beta_1 \\ \beta_2 \\ \vdots \\ \beta_p \end{bmatrix} \\
&= x_1\beta_1 + x_2\beta_2 + \dots + x_p\beta_p
\end{aligned}$$

**Theorem 0.7** (Matrix multiplication is associative). *For an  $m \times k$  matrix  $\mathbf{A}$ , a  $k \times l$  matrix  $\mathbf{B}$ , and an  $l \times n$  matrix  $\mathbf{C}$ :*

$$\underbrace{(\mathbf{AB})\mathbf{C}}_{m \times n} = \underbrace{\mathbf{A}(\mathbf{BC})}_{m \times n}$$

### **i** Proof

*Proof.* Entry  $(i, j)$  of each side, from Definition 0.20:

$$\begin{aligned}
[(\mathbf{AB})\mathbf{C}]_{ij} &= \sum_{t=1}^l (\mathbf{AB})_{it} c_{tj} && \text{(definition of } (\mathbf{AB})\mathbf{C} \text{)} \\
&= \sum_{t=1}^l \left( \sum_{s=1}^k a_{is} b_{st} \right) c_{tj} && \text{(definition of } \mathbf{AB} \text{)} \\
&= \sum_{s=1}^k a_{is} \left( \sum_{t=1}^l b_{st} c_{tj} \right) && \text{(distribute and swap the finite sums)} \\
&= \sum_{s=1}^k a_{is} (\mathbf{BC})_{sj} && \text{(definition of } \mathbf{BC} \text{)} \\
&= [\mathbf{A}(\mathbf{BC})]_{ij} && \text{(definition of } \mathbf{A}(\mathbf{BC}) \text{)}
\end{aligned}$$

□

**Theorem 0.8** (Matrix multiplication is distributive over addition). *For an  $m \times k$  matrix  $\mathbf{A}$  and  $k \times n$  matrices  $\mathbf{B}$  and  $\mathbf{C}$ :*

$$\mathbf{A}(\mathbf{B} + \mathbf{C}) = \mathbf{AB} + \mathbf{AC}$$

*For  $m \times k$  matrices  $\mathbf{A}$  and  $\mathbf{B}$  and a  $k \times n$  matrix  $\mathbf{C}$ :*

$$(\mathbf{A} + \mathbf{B})\mathbf{C} = \mathbf{AC} + \mathbf{BC}$$

**Exercise 0.4** (Which of these products are defined?). Let  $\mathbf{A}$  be  $4 \times 3$ , let  $\mathbf{B}$  be  $3 \times 2$ , and let  $\tilde{x} \in \mathbb{R}^3$ . For each expression, say whether it is defined, and if so give the shape of the result:

1.  $\mathbf{AB}$
2.  $\mathbf{BA}$
3.  $\mathbf{A}\tilde{x}$
4.  $\tilde{x}^\top \tilde{x}$
5.  $\tilde{x}\tilde{x}^\top$

#### Solution

*Solution 0.4.* A vector in  $\mathbb{R}^3$  is a  $3 \times 1$  matrix, so the same rule settles every case.

1.  $\mathbf{AB}$ :  $(4 \times 3)(3 \times 2)$ . The inner pair is 3 and 3, so it is defined, and the result is  $4 \times 2$ .
2.  $\mathbf{BA}$ :  $(3 \times 2)(4 \times 3)$ . The inner pair is 2 and 4, which do not match, so it is **not defined**. Matrix multiplication is not commutative, and this is the blunt form of that: swapping the order can leave an expression that means nothing.
3.  $\mathbf{A}\tilde{x}$ :  $(4 \times 3)(3 \times 1)$ , defined, result  $4 \times 1$  — a vector in  $\mathbb{R}^4$ .
4.  $\tilde{x}^\top \tilde{x}$ :  $(1 \times 3)(3 \times 1)$ , defined, result  $1 \times 1$  — a single number. The product is the inner product of Equation 1.
5.  $\tilde{x}\tilde{x}^\top$ :  $(3 \times 1)(1 \times 3)$ , defined, result  $3 \times 3$  — a matrix.

The last two use the same two vectors and differ only in order, and they return objects of different kinds. Reading a transpose as decoration rather than as a shape change is the commonest way to lose track of an expression.

### Matrix-vector multiplication

**Definition 0.21** (Matrix-vector multiplication). The **matrix-vector product** of an  $m \times p$  matrix  $\mathbf{A}$  and a  $p \times 1$  column vector  $\tilde{x}$  is the  $m \times 1$  column vector  $\mathbf{A}\tilde{x}$  with entries:

$$(\mathbf{A}\tilde{x})_i = \sum_{j=1}^p a_{ij} x_j$$

*Remark 0.11* (Each entry is a dot product). Matrix-vector multiplication extends the dot product (Definition 0.5): entry  $i$  of  $\mathbf{A}\tilde{x}$  is the dot product of row  $i$  of  $\mathbf{A}$  with  $\tilde{x}$ . For example,

$$\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \begin{bmatrix} 1 \\ -1 \end{bmatrix} = \begin{bmatrix} 1 \cdot 1 + 2 \cdot (-1) \\ 3 \cdot 1 + 4 \cdot (-1) \end{bmatrix} = \begin{bmatrix} -1 \\ -1 \end{bmatrix}.$$

## Linear and affine maps

**Definition 0.22** (Linear map). A function  $f : \mathbb{R}^p \rightarrow \mathbb{R}^m$  is a **linear map** (or linear transformation) if it preserves vector addition and scalar multiplication: for all  $\tilde{x}, \tilde{y} \in \mathbb{R}^p$  and  $c \in \mathbb{R}$ ,

1.  $f(\tilde{x} + \tilde{y}) = f(\tilde{x}) + f(\tilde{y})$
2.  $f(c\tilde{x}) = cf(\tilde{x})$

Every such function can be represented as multiplication by an  $m \times p$  matrix  $\mathbf{A}$ :

$$f(\tilde{x}) = \mathbf{A}\tilde{x}$$

A linear map always maps the zero vector to the zero vector:  $f(\tilde{0}) = \tilde{0}$ .

**Definition 0.23** (Affine transformation). An **affine transformation** is a function  $f : \mathbb{R}^p \rightarrow \mathbb{R}^m$  composed of a linear map followed by a translation:

$$f(\tilde{x}) = \mathbf{A}\tilde{x} + \tilde{b}, \quad \mathbf{A} \in \mathbb{R}^{m \times p}, \quad \tilde{b} \in \mathbb{R}^m$$

For a scalar-valued function ( $m = 1$ ), this is an inner product plus an offset:

$$f(\tilde{x}) = \tilde{w}^\top \tilde{x} + b, \quad \tilde{w} \in \mathbb{R}^p, \quad b \in \mathbb{R} \quad (4)$$

When  $\tilde{b} \neq \tilde{0}$  (or  $b \neq 0$ ), an affine transformation does not send  $\tilde{0}$  to  $\tilde{0}$ , so it is not strictly linear. In machine learning and statistics, functions of this form are commonly referred to as **linear models** or linear functions, with  $\tilde{b}$  (or  $b$ ) called the **bias** or **intercept**.

**Definition 0.24** (Hyperplane). A **hyperplane** in  $\mathbb{R}^p$  is an affine subspace of dimension  $p - 1$ . It is defined as the set of points  $\tilde{x} \in \mathbb{R}^p$  satisfying:

$$\tilde{w}^\top \tilde{x} + b = 0$$

where  $\tilde{w} \in \mathbb{R}^p \setminus \{\tilde{0}\}$  is the normal vector orthogonal to the hyperplane and  $b \in \mathbb{R}$  is the offset. When  $b = 0$ , the hyperplane passes through the origin and forms a  $(p - 1)$ -dimensional linear subspace.

Geometrically:

- In  $\mathbb{R}^2$  ( $p = 2$ ), a hyperplane is a 1-dimensional line:  $w_1x_1 + w_2x_2 + b = 0$ .
- In  $\mathbb{R}^3$  ( $p = 3$ ), a hyperplane is a 2-dimensional plane.
- A hyperplane divides the space  $\mathbb{R}^p$  into two open **half-spaces**:  $\{\tilde{x} : \tilde{w}^\top \tilde{x} + b > 0\}$  and  $\{\tilde{x} : \tilde{w}^\top \tilde{x} + b < 0\}$ .

The course website's version of this page has an interactive plot here.

**Exercise 0.5** (Write a rule in matrix form). A clinic scores a patient's risk from three measurements.

| symbol | meaning                 | units             |
|--------|-------------------------|-------------------|
| $x_1$  | age                     | years             |
| $x_2$  | systolic blood pressure | mmHg              |
| $x_3$  | body mass index         | kg/m <sup>2</sup> |

The rule in use is

$$\text{score} = 0.4x_1 + 0.2x_2 + 1.5x_3 - 30$$

Write this rule in the form of Equation 4, naming  $\tilde{w}$ ,  $b$  and  $p$  explicitly, and give the score for a patient aged 50 with blood pressure 130 and BMI 28.

### Solution

*Solution 0.5.* Read the coefficients straight off the rule. There are three inputs, so  $p = 3$ :

$$\tilde{w} = \begin{bmatrix} 0.4 \\ 0.2 \\ 1.5 \end{bmatrix}, \quad b = -30$$

and the rule is  $f(\tilde{x}) = \tilde{w}^\top \tilde{x} + b$ . For the patient,

$$\tilde{x} = \begin{bmatrix} 50 \\ 130 \\ 28 \end{bmatrix}$$

$$\tilde{w}^\top \tilde{x} = (0.4)(50) + (0.2)(130) + (1.5)(28) = 20 + 26 + 42 = 88$$

$$f(\tilde{x}) = 88 - 30 = 58$$

Two things are worth noticing. The coefficients are not comparable as they stand: 1.5 is the largest number but BMI is measured on the smallest scale, so the weight on a feature

says nothing on its own about how much that feature matters. And  $b = -30$  is doing real work — without it the score would be 88, which is a different clinical claim entirely.

**Exercise 0.6** (Which of these are linear in  $\tilde{w}$ ?). Each of the five expressions is a function of the weights  $\tilde{w} \in \mathbb{R}^p$ , with the data  $\tilde{x}$  and  $y$  held fixed. Say which ones have the form Equation 4 — an inner product plus an offset — and which do not.

1.  $\tilde{w}^\top \tilde{x}$
2.  $\tilde{w}^\top \tilde{x} - y$
3.  $(\tilde{w}^\top \tilde{x} - y)^2$
4.  $\|\tilde{w}\|_2^2$
5.  $w_1 x_1 + 3$

### Solution

*Solution 0.6.*

1. **Linear.** It is Equation 4 with  $b = 0$ .
2. **Linear.** Still an inner product plus an offset; here the offset is  $b = -y$ , which is a constant because  $y$  is held fixed.
3. **Not linear.** Squaring is not an inner product plus an offset. This expression is the squared error of one observation, and its curvature in  $\tilde{w}$  is exactly why fitting a model is an optimization problem rather than a lookup.
4. **Not linear.** By Equation 2 this is  $\tilde{w}^\top \tilde{w}$ , which has  $\tilde{w}$  in both slots at once. Hold either slot fixed and it is linear in the other; as a function of the single variable  $\tilde{w}$  it is quadratic.
5. **Linear.** It is Equation 4 with coefficient vector  $\tilde{a} = (x_1, 0, \dots, 0)^\top$  and  $b = 3$ . The data  $\tilde{x}$  is untouched —  $\tilde{a}$  is a new vector built from its first entry.

Items 1, 2 and 5 are linear and items 3 and 4 are not, which is the split that matters: a *model* can be linear in its weights while the *quantity we minimize* is not. Squared error and the norm penalty are both curved in  $\tilde{w}$ , and that curvature is what makes fitting an optimization problem at all: there is always a downhill direction to follow. Curvature alone does not promise a *single* answer. The squared error of one observation is completely flat along a whole line of weight vectors, every one of which fits that observation exactly, and pinning down one of them is part of what the norm penalty is added for in week 3.

## Transposes of sums and products

**Theorem 0.9** (Transpose of a sum). *For  $m \times n$  matrices  $\mathbf{A}$  and  $\mathbf{B}$ :*

$$(\mathbf{A} + \mathbf{B})^\top = \mathbf{A}^\top + \mathbf{B}^\top$$

*In particular, for column vectors  $\tilde{x}$  and  $\tilde{y}$  of the same length:*

$$(\tilde{x} + \tilde{y})^\top = \tilde{x}^\top + \tilde{y}^\top$$

### Proof

*Proof.* Entry  $(i, j)$  of each side:

$$\begin{aligned} [(\mathbf{A} + \mathbf{B})^\top]_{ij} &= (\mathbf{A} + \mathbf{B})_{ji} && \text{(definition of the transpose)} \\ &= a_{ji} + b_{ji} && \text{(definition of matrix addition)} \\ &= (\mathbf{A}^\top)_{ij} + (\mathbf{B}^\top)_{ij} && \text{(definition of the transpose)} \\ &= [\mathbf{A}^\top + \mathbf{B}^\top]_{ij} && \text{(definition of matrix addition)} \end{aligned}$$

□

**Theorem 0.10** (Transpose of a product). *For an  $m \times k$  matrix  $\mathbf{A}$  and a  $k \times n$  matrix  $\mathbf{B}$ :*

$$\underbrace{(\mathbf{AB})^\top}_{n \times m} = \underbrace{\mathbf{B}^\top}_{n \times k} \underbrace{\mathbf{A}^\top}_{k \times m}$$

### Proof

*Proof.* Entry  $(i, j)$  of each side:

$$\begin{aligned} [(\mathbf{AB})^\top]_{ij} &= (\mathbf{AB})_{ji} && \text{(definition of the transpose)} \\ &= \sum_{s=1}^k a_{js} b_{si} && \text{(definition of matrix multiplication)} \\ &= \sum_{s=1}^k (\mathbf{B}^\top)_{is} (\mathbf{A}^\top)_{sj} && \text{(definition of the transpose; reorder each product)} \\ &= [\mathbf{B}^\top \mathbf{A}^\top]_{ij} && \text{(definition of matrix multiplication)} \end{aligned}$$

□

*Remark 0.12* (The order of the factors reverses). Transposing a product reverses the order of the factors. Keeping the original order usually gives an undefined product or a different matrix. For example, with  $\mathbf{A} = [1, 2]$  ( $1 \times 2$ ) and  $\mathbf{B} = \begin{bmatrix} 3 \\ 4 \end{bmatrix}$  ( $2 \times 1$ ),  $\mathbf{AB} = 1 \cdot 3 + 2 \cdot 4 = 11$ , so  $(\mathbf{AB})^\top = 11$ , and

$$\mathbf{B}^\top \mathbf{A}^\top = [3, 4] \begin{bmatrix} 1 \\ 2 \end{bmatrix} = 11, \quad \mathbf{A}^\top \mathbf{B}^\top = \begin{bmatrix} 1 \\ 2 \end{bmatrix} [3, 4] = \begin{bmatrix} 3 & 4 \\ 6 & 8 \end{bmatrix}.$$

If instead  $\mathbf{A}$  is  $2 \times 3$  and  $\mathbf{B}$  is  $3 \times 1$ , then  $\mathbf{B}^\top \mathbf{A}^\top$  is a  $(1 \times 3)(3 \times 2)$  product, but  $\mathbf{A}^\top \mathbf{B}^\top$  is a  $(3 \times 2)(1 \times 3)$  product, which is undefined.

## Rank

**Definition 0.25** (Linearly independent vectors). Vectors  $\tilde{v}_1, \dots, \tilde{v}_k$  of the same length  $p$  are **linearly independent** if the only numbers  $c_1, \dots, c_k$  with

$$c_1 \tilde{v}_1 + \dots + c_k \tilde{v}_k = \tilde{0}$$

are  $c_1 = \dots = c_k = 0$ .

**Example 0.8** (Independent and dependent pairs of vectors).

- $\tilde{v}_1 = (1, 0)$  and  $\tilde{v}_2 = (1, 1)$  are linearly independent:  $c_1 \tilde{v}_1 + c_2 \tilde{v}_2 = (c_1 + c_2, c_2)$ , which is  $\tilde{0}$  only if  $c_2 = 0$  and then  $c_1 = 0$ .
- $\tilde{v}_1 = (1, 2)$  and  $\tilde{v}_2 = (2, 4)$  are not:  $2\tilde{v}_1 - \tilde{v}_2 = \tilde{0}$ .

**Definition 0.26** (Rank). The **rank** of a matrix  $\mathbf{A}$ , written  $\text{rank}(\mathbf{A})$ , is the largest number of columns of  $\mathbf{A}$  that are linearly independent (Definition 0.25).

**Example 0.9** (The rank of two  $3 \times 2$  matrices). The matrix  $\begin{bmatrix} 1 & 1 \\ 1 & 2 \\ 1 & 3 \end{bmatrix}$  has rank 2: if

$c_1(1, 1, 1) + c_2(1, 2, 3) = \tilde{0}$ , then subtracting the first entry from the second gives  $c_2 = 0$ , and then  $c_1 = 0$ .

The matrix  $\begin{bmatrix} 1 & 2 \\ 1 & 2 \\ 1 & 2 \end{bmatrix}$  has rank 1: its second column is twice its first, so the two columns are not linearly independent, but the first column on its own is.

**Definition 0.27** (Full column rank). An  $n \times p$  matrix has **full column rank** if its rank (Definition 0.26) is  $p$ , so that all of its columns are linearly independent.

*Remark 0.13* (Full column rank needs at least as many rows as columns). In Example 0.9, the first matrix has full column rank: its rank is 2, and it has 2 columns. The second does not, because its rank is 1.

An  $n \times p$  matrix can have full column rank only when  $p \leq n$ , because more than  $n$  vectors of length  $n$  are never linearly independent. For example, the  $2 \times 3$  matrix  $\begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \end{bmatrix}$  has 3 columns of length 2, and  $1 \cdot (1, 0) + 1 \cdot (0, 1) - 1 \cdot (1, 1) = \tilde{0}$ , with coefficients that are not all zero. So its 3 columns are not linearly independent, and its rank is less than 3.

## Outer product

**Definition 0.28** (Outer product). The **outer product** of a vector  $\tilde{u}$  of length  $m$  and a vector  $\tilde{v}$  of length  $n$  is the  $m \times n$  matrix product (Definition 0.20)  $\tilde{u} \tilde{v}^\top$  of the column vector  $\tilde{u}$  with the row vector  $\tilde{v}^\top$ . Its entries are

$$\left( \begin{array}{c|c} \tilde{u} & \tilde{v}^\top \\ \hline m \times 1 & 1 \times n \end{array} \right)_{ij} = u_i v_j \quad (\text{definition of matrix multiplication; the sum has one term})$$

**Example 0.10** (An outer product). For  $\tilde{u} = (1, 2, 3)$  and  $\tilde{v} = (4, 5)$ :

$$\begin{aligned} \tilde{u} \tilde{v}^\top &= \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} \begin{bmatrix} 4 & 5 \end{bmatrix} && (\text{definition of the outer product}) \\ &= \begin{bmatrix} 1 \cdot 4 & 1 \cdot 5 \\ 2 \cdot 4 & 2 \cdot 5 \\ 3 \cdot 4 & 3 \cdot 5 \end{bmatrix} && (\text{entry } (i, j) \text{ is } u_i v_j) \\ &= \begin{bmatrix} 4 & 5 \\ 8 & 10 \\ 12 & 15 \end{bmatrix} && (\text{multiply}) \end{aligned}$$

*Remark 0.14* (Outer product and dot product). The dot product  $\tilde{u}^\top \tilde{v}$  (Example 0.7) needs two vectors of the same length and gives a number. The outer product  $\tilde{u} \tilde{v}^\top$  takes vectors of any two lengths and gives a matrix (Banerjee and Roy 2014, chap. 1, p. 11).

For example,  $\tilde{u} = (1, 2, 3)$  and  $\tilde{v} = (4, 5)$  in Example 0.10 have different lengths, so they have no dot product, but their outer product is a  $3 \times 2$  matrix. For  $\tilde{a} = (1, 2)$  and  $\tilde{b} = (3, 4)$ , which have the same length, both products exist:

$$\tilde{a}^\top \tilde{b} = 1 \cdot 3 + 2 \cdot 4 = 11 \quad \tilde{a} \tilde{b}^\top = \begin{bmatrix} 1 \cdot 3 & 1 \cdot 4 \\ 2 \cdot 3 & 2 \cdot 4 \end{bmatrix} = \begin{bmatrix} 3 & 4 \\ 6 & 8 \end{bmatrix}$$

**Theorem 0.11** (An outer product has rank one). *If  $\tilde{u}$  is a nonzero vector of length  $m$  and  $\tilde{v}$  is a nonzero vector of length  $n$ , then  $\text{rank}(\tilde{u} \tilde{v}^\top) = 1$  (Definition 0.26).*

### **i** Proof

*Proof.* Column  $j$  of  $\tilde{u} \tilde{v}^\top$  has entries  $u_1 v_j, \dots, u_m v_j$ , so it is the vector  $v_j \tilde{u}$ .

**The rank is at least one.** Because  $\tilde{v} \neq \tilde{0}$ , some entry  $v_j$  is nonzero. Then column  $j$ ,  $v_j \tilde{u}$ , is a nonzero vector, and a single nonzero vector is linearly independent (Definition 0.25):  $c(v_j \tilde{u}) = \tilde{0}$  forces  $c = 0$ .

**The rank is at most one.** Take any two columns  $j \neq k$ . If  $v_j = v_k = 0$ , both columns are  $\tilde{0}$ , and  $1 \cdot (v_j \tilde{u}) + 1 \cdot (v_k \tilde{u}) = \tilde{0}$ . Otherwise, use the coefficients  $c_j = v_k$  and  $c_k = -v_j$ , which are not both zero:

$$\begin{aligned} v_k(v_j \tilde{u}) - v_j(v_k \tilde{u}) &= (v_k v_j - v_j v_k) \tilde{u} && \text{(collect the scalar multiples of } \tilde{u}) \\ &= 0 \cdot \tilde{u} && \text{(multiplication of numbers is commutative)} \\ &= \tilde{0} && \text{(a zero multiple of a vector is } \tilde{0}) \end{aligned}$$

Either way, every pair of columns has a combination with coefficients that are not all zero and that equals  $\tilde{0}$ . Now take any set of two or more columns. It contains a pair  $j \neq k$ . Use that pair's coefficients for columns  $j$  and  $k$ , and the coefficient 0 for every other column in the set. This combination equals  $\tilde{0}$ , and its coefficients are not all zero, so the set is not linearly independent (Definition 0.25). The largest linearly independent set of columns therefore has one column.  $\square$

**Example 0.11** (The rank of an outer product). In Example 0.10, the second column  $(5, 10, 15)$  of  $\tilde{u} \tilde{v}^\top$  is  $\frac{5}{4}$  times the first column  $(4, 8, 12)$ , because the columns are  $v_1 \tilde{u} = 4\tilde{u}$  and  $v_2 \tilde{u} = 5\tilde{u}$ . So the two columns are not linearly independent, and the  $3 \times 2$  matrix has rank 1.

## Special Matrices

One special matrix, the zero matrix (Definition 0.17), appeared earlier, in the section on matrix addition.

**Definition 0.29** (Square matrix). A matrix is **square** if it has the same number of rows as columns.

**Definition 0.30** (Order of a square matrix). The **order** of a square matrix (Definition 0.29) is its number of rows, which equals its number of columns.

**Definition 0.31** (Matrix power). For a square matrix  $\mathbf{A}$  of order  $p$  and a positive integer  $k$ , the  $k$ -th **power** of  $\mathbf{A}$  is:

$$\mathbf{A}^k = \underbrace{\mathbf{A} \mathbf{A} \cdots \mathbf{A}}_{k \text{ copies}}$$

In particular,  $\mathbf{A}^2 = \mathbf{A}\mathbf{A}$ .

**Definition 0.32** (Identity matrix). The  $p \times p$  **identity matrix**  $\mathbf{I}_p$  (or  $\mathbf{I}$  when the size is clear from context) has ones on the main diagonal and zeros elsewhere:

$$(\mathbf{I}_p)_{ij} = \begin{cases} 1 & \text{if } i = j \\ 0 & \text{if } i \neq j \end{cases} \quad \mathbf{I}_p = \begin{bmatrix} 1 & 0 & \cdots & 0 \\ 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 1 \end{bmatrix}$$

Equivalently, the entries of the identity matrix are given by the **Kronecker delta**:  $(\mathbf{I}_p)_{ij} = \delta_{ij}$ .

**Theorem 0.12** (Identity matrix is a multiplicative identity). *For any  $m \times p$  matrix  $\mathbf{A}$ :*

$$\mathbf{A} \mathbf{I}_p = \mathbf{A}$$

$$\mathbf{I}_m \mathbf{A} = \mathbf{A}$$

**Definition 0.33** (Symmetric matrix). A square matrix  $\mathbf{A}$  is **symmetric** if  $\mathbf{A}^\top = \mathbf{A}$ , i.e.,  $a_{ij} = a_{ji}$  for all  $i$  and  $j$ .

*Remark 0.15* (Symmetric matrices in statistics). Covariance matrices are symmetric: entry  $(i, j)$  of the covariance matrix of a random vector  $\tilde{Y}$  is  $\text{Cov}(Y_i, Y_j)$ , and  $\text{Cov}(Y_i, Y_j) = \text{Cov}(Y_j, Y_i)$ . For example, if  $\text{Var}(Y_1) = 4$ ,  $\text{Var}(Y_2) = 9$ , and  $\text{Cov}(Y_1, Y_2) = 1$ , the covariance matrix of  $(Y_1, Y_2)$  is

$$\begin{bmatrix} 4 & 1 \\ 1 & 9 \end{bmatrix}.$$

**Definition 0.34** (Diagonal matrix). A square matrix  $\mathbf{D}$  is a **diagonal matrix** if all off-diagonal entries are zero:  $d_{ij} = 0$  whenever  $i \neq j$ :

$$\mathbf{D} = \begin{bmatrix} d_1 & 0 & \cdots & 0 \\ 0 & d_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & d_p \end{bmatrix}$$

*Remark 0.16* (diag notation). For numbers  $d_1, \dots, d_p$ ,  $\text{diag}(d_1, \dots, d_p)$  is the  $p \times p$  diagonal matrix with diagonal entries  $d_1, \dots, d_p$ . For example,

$$\text{diag}(2, 5) = \begin{bmatrix} 2 & 0 \\ 0 & 5 \end{bmatrix},$$

and the  $p \times p$  identity matrix is  $\mathbf{I}_p = \text{diag}(1, \dots, 1)$ .

**Definition 0.35** (Matrix inverse). For a square  $p \times p$  matrix  $\mathbf{A}$ , the **inverse**  $\mathbf{A}^{-1}$  (if it exists) is the unique matrix satisfying:

$$\mathbf{A} \mathbf{A}^{-1} = \mathbf{A}^{-1} \mathbf{A} = \mathbf{I}_p$$

**Definition 0.36** (Invertible matrix). A  $p \times p$  matrix  $\mathbf{A}$  is **invertible** (or *non-singular*) if some  $p \times p$  matrix  $\mathbf{B}$  satisfies

$$\mathbf{AB} = \mathbf{BA} = \mathbf{I}_p,$$

where  $\mathbf{I}_p$  is the identity matrix (Definition 0.32). A square matrix that is not invertible is *singular*.

**Example 0.12** (An invertible matrix and a singular one). The matrix  $\mathbf{A} = \begin{bmatrix} 2 & 1 \\ 0 & 1 \end{bmatrix}$  is

invertible, with  $\mathbf{A}^{-1} = \begin{bmatrix} 0.5 & -0.5 \\ 0 & 1 \end{bmatrix}$ : multiplying out gives  $\mathbf{A}\mathbf{A}^{-1} = \mathbf{A}^{-1}\mathbf{A} = \mathbf{I}_2$ .

The matrix  $\mathbf{M} = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$  is singular: for any  $2 \times 2$  matrix  $\mathbf{C}$ , the two rows of  $\mathbf{MC}$  are equal, so  $\mathbf{MC}$  can never be  $\mathbf{I}_2$ , whose two rows differ.

*Remark 0.17* (Invertible matrices and inverses). Let  $\mathbf{A}$  be a  $p \times p$  matrix, and suppose that two  $p \times p$  matrices  $\mathbf{B}$  and  $\mathbf{C}$  satisfy  $\mathbf{AB} = \mathbf{BA} = \mathbf{I}_p$  and  $\mathbf{AC} = \mathbf{CA} = \mathbf{I}_p$ . Then

$$\begin{aligned} \mathbf{B} &= \mathbf{BI}_p && \text{(identity matrix)} \\ &= \mathbf{B}(\mathbf{AC}) && (\mathbf{AC} = \mathbf{I}_p) \\ &= (\mathbf{BA})\mathbf{C} && \text{(matrix multiplication is associative)} \\ &= \mathbf{I}_p\mathbf{C} && (\mathbf{BA} = \mathbf{I}_p) \\ &= \mathbf{C} && \text{(identity matrix)} \end{aligned}$$

The steps use Theorem 0.7 and Theorem 0.12. Since  $\mathbf{B} = \mathbf{C}$ , at most one matrix satisfies these equations, which is the uniqueness that Definition 0.35 asserts. When  $\mathbf{A}$  is invertible, the matrix  $\mathbf{B}$  in Definition 0.36 is therefore the inverse  $\mathbf{A}^{-1}$  of  $\mathbf{A}$ , and  $\mathbf{A}$  is invertible exactly when it has an inverse. For example, for  $\mathbf{A} = \begin{bmatrix} 2 & 1 \\ 0 & 1 \end{bmatrix}$  in Example 0.12,  $\begin{bmatrix} 0.5 & -0.5 \\ 0 & 1 \end{bmatrix}$  is the only  $2 \times 2$  matrix  $\mathbf{B}$  with  $\mathbf{AB} = \mathbf{BA} = \mathbf{I}_2$ , so it is  $\mathbf{A}^{-1}$ .

**Theorem 0.13** (Inverse of a product). *If  $\mathbf{A}$  and  $\mathbf{B}$  are invertible  $p \times p$  matrices, then  $\mathbf{AB}$  is invertible, and*

$$(\mathbf{AB})^{-1} = \mathbf{B}^{-1}\mathbf{A}^{-1}$$

### Proof

*Proof.* Multiply  $\mathbf{AB}$  by the candidate inverse on the right:

$$\begin{aligned} (\mathbf{AB})(\mathbf{B}^{-1}\mathbf{A}^{-1}) &= \mathbf{A}(\mathbf{BB}^{-1})\mathbf{A}^{-1} && \text{(matrix multiplication is associative)} \\ &= \mathbf{AI}_p\mathbf{A}^{-1} && \text{(definition of } \mathbf{B}^{-1}) \\ &= \mathbf{AA}^{-1} && \text{(identity matrix)} \\ &= \mathbf{I}_p && \text{(definition of } \mathbf{A}^{-1}) \end{aligned}$$

and on the left:

$$\begin{aligned}
(\mathbf{B}^{-1}\mathbf{A}^{-1})(\mathbf{AB}) &= \mathbf{B}^{-1}(\mathbf{A}^{-1}\mathbf{A})\mathbf{B} && \text{(matrix multiplication is associative)} \\
&= \mathbf{B}^{-1}\mathbf{I}_p\mathbf{B} && \text{(definition of } \mathbf{A}^{-1} \text{)} \\
&= \mathbf{B}^{-1}\mathbf{B} && \text{(identity matrix)} \\
&= \mathbf{I}_p && \text{(definition of } \mathbf{B}^{-1} \text{)}
\end{aligned}$$

So  $\mathbf{B}^{-1}\mathbf{A}^{-1}$  satisfies Definition 0.35 for  $\mathbf{AB}$ . The steps use Theorem 0.7 and Theorem 0.12.  $\square$

**Theorem 0.14** (Inverse of a transpose). *If  $\mathbf{A}$  is an invertible  $p \times p$  matrix, then  $\mathbf{A}^\top$  is invertible, and*

$$(\mathbf{A}^\top)^{-1} = (\mathbf{A}^{-1})^\top$$

### **i** Proof

*Proof.* Multiply  $\mathbf{A}^\top$  by the candidate inverse on each side:

$$\begin{aligned}
\mathbf{A}^\top (\mathbf{A}^{-1})^\top &= (\mathbf{A}^{-1}\mathbf{A})^\top && \text{(transpose of a product)} \\
&= \mathbf{I}_p^\top && \text{(definition of } \mathbf{A}^{-1} \text{)} \\
&= \mathbf{I}_p && (\mathbf{I}_p \text{ is symmetric)} \\
(\mathbf{A}^{-1})^\top \mathbf{A}^\top &= (\mathbf{AA}^{-1})^\top && \text{(transpose of a product)} \\
&= \mathbf{I}_p^\top && \text{(definition of } \mathbf{A}^{-1} \text{)} \\
&= \mathbf{I}_p && (\mathbf{I}_p \text{ is symmetric)}
\end{aligned}$$

So  $(\mathbf{A}^{-1})^\top$  satisfies Definition 0.35 for  $\mathbf{A}^\top$ . The first step of each display is Theorem 0.10.  $\square$

**Example 0.13** (Inverting a transpose). For  $\mathbf{A} = \begin{bmatrix} 2 & 1 \\ 0 & 1 \end{bmatrix}$ , Example 0.12 gives  $\mathbf{A}^{-1} = \begin{bmatrix} 0.5 & -0.5 \\ 0 & 1 \end{bmatrix}$ , so Theorem 0.14 says

$$(\mathbf{A}^\top)^{-1} = \left( \begin{bmatrix} 2 & 0 \\ 1 & 1 \end{bmatrix} \right)^{-1} = (\mathbf{A}^{-1})^\top = \begin{bmatrix} 0.5 & 0 \\ -0.5 & 1 \end{bmatrix},$$

and multiplying  $\begin{bmatrix} 2 & 0 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 0.5 & 0 \\ -0.5 & 1 \end{bmatrix}$  out does give  $\mathbf{I}_2$ .

**Corollary 0.1** (The inverse of a symmetric matrix is symmetric). *If  $\mathbf{A}$  is symmetric and invertible, then  $\mathbf{A}^{-1}$  is symmetric.*

**i** Proof

*Proof.*

$$\begin{aligned} (\mathbf{A}^{-1})^\top &= (\mathbf{A}^\top)^{-1} \quad (\text{inverse of a transpose}) \\ &= \mathbf{A}^{-1} \quad (\mathbf{A} \text{ is symmetric}) \end{aligned}$$

The first step is Theorem 0.14. □

**Example 0.14** (Inverting a symmetric matrix).  $\mathbf{S} = \begin{bmatrix} 2 & 1 \\ 1 & 1 \end{bmatrix}$  is symmetric, and  $\mathbf{S}^{-1} = \begin{bmatrix} 1 & -1 \\ -1 & 2 \end{bmatrix}$  is symmetric too, as Corollary 0.1 says.

**Definition 0.37** (Idempotent matrix). A square matrix  $\mathbf{A}$  is **idempotent** if

$$\mathbf{A}^2 = \mathbf{A}$$

**Definition 0.38** (Orthogonal projection matrix). A square matrix  $\mathbf{P}$  is an **orthogonal projection matrix** if it is both symmetric (Definition 0.33) and idempotent (Definition 0.37):

$$\mathbf{P}^\top = \mathbf{P} \quad \text{and} \quad \mathbf{P}^2 = \mathbf{P}$$

**Definition 0.39** (Oblique projection). A square matrix is an **oblique projection** if it is idempotent (Definition 0.37) but not symmetric (Definition 0.33).

**Example 0.15** (An orthogonal projection and an oblique one).  $\mathbf{P} = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$  is symmetric, and  $\mathbf{P}^2 = \mathbf{P}$ , so  $\mathbf{P}$  is an orthogonal projection matrix; it maps  $(v_1, v_2)$  to  $(v_1, 0)$ .  $\mathbf{Q} = \begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix}$  is idempotent:

$$\mathbf{Q}^2 = \begin{bmatrix} 1 \cdot 1 + 1 \cdot 0 & 1 \cdot 1 + 1 \cdot 0 \\ 0 \cdot 1 + 0 \cdot 0 & 0 \cdot 1 + 0 \cdot 0 \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix} = \mathbf{Q},$$

but  $\mathbf{Q}^\top \neq \mathbf{Q}$ , so  $\mathbf{Q}$  is an oblique projection, not an orthogonal projection matrix.

*Remark 0.18* (What “projection matrix” means in these notes). Some texts call any idempotent matrix a projection matrix, so that both  $\mathbf{P}$  and  $\mathbf{Q}$  in Example 0.15 would count as one. Regression texts often say “projection matrix” when they mean an orthogonal one. In these notes, “projection matrix” always means an orthogonal projection matrix in the sense of Definition 0.38, such as  $\mathbf{P}$  in Example 0.15, and never an oblique projection (Definition 0.39) such as  $\mathbf{Q}$ .

**Theorem 0.15** (Complement of a projection matrix). *If  $\mathbf{P}$  is a  $p \times p$  orthogonal projection matrix (Definition 0.38), then  $\mathbf{I}_p - \mathbf{P}$  is also an orthogonal projection matrix.*

### **i** Proof

*Proof.* We verify symmetry and idempotency.

**Symmetry:**

$$(\mathbf{I} - \mathbf{P})^\top = \mathbf{I}^\top - \mathbf{P}^\top = \mathbf{I} - \mathbf{P}$$

**Idempotency:**

$$\begin{aligned} (\mathbf{I} - \mathbf{P})^2 &= (\mathbf{I} - \mathbf{P})(\mathbf{I} - \mathbf{P}) \\ &= \mathbf{I} - \mathbf{P} - \mathbf{P} + \mathbf{P}^2 \\ &= \mathbf{I} - \mathbf{P} - \mathbf{P} + \mathbf{P} \\ &= \mathbf{I} - \mathbf{P} \end{aligned}$$

□

**Theorem 0.16** (Projection matrices produce orthogonal decompositions). *If  $\mathbf{P}$  is a  $p \times p$  orthogonal projection matrix (Definition 0.38) and  $\tilde{v}$  is any vector of length  $p$ , then the two components of the decomposition*

$$\tilde{v} = \underbrace{\mathbf{P}\tilde{v}}_{\text{projected}} + \underbrace{(\mathbf{I}_p - \mathbf{P})\tilde{v}}_{\text{residual}}$$

*are orthogonal:*

$$\mathbf{P}\tilde{v} \perp (\mathbf{I}_p - \mathbf{P})\tilde{v}$$

**i** Proof

*Proof.*

$$\begin{aligned}
 (\mathbf{P}\tilde{v})^\top (\mathbf{I} - \mathbf{P})\tilde{v} &= \tilde{v}^\top \mathbf{P}^\top (\mathbf{I} - \mathbf{P})\tilde{v} \\
 &= \tilde{v}^\top \mathbf{P} (\mathbf{I} - \mathbf{P})\tilde{v} \\
 &= \tilde{v}^\top (\mathbf{P} - \mathbf{P}^2)\tilde{v} \\
 &= \tilde{v}^\top (\mathbf{P} - \mathbf{P})\tilde{v} \\
 &= \tilde{v}^\top \mathbf{0} \tilde{v} \\
 &= 0
 \end{aligned}$$

where the second line uses symmetry ( $\mathbf{P}^\top = \mathbf{P}$ ) and the fourth line uses idempotency ( $\mathbf{P}^2 = \mathbf{P}$ ).  $\square$

**Definition 0.40** (Orthogonal matrix). A  $p \times p$  matrix  $\mathbf{Q}$  is **orthogonal** if

$$\underbrace{\mathbf{Q}^\top}_{p \times p} \underbrace{\mathbf{Q}}_{p \times p} = \mathbf{I}_p$$

**Example 0.16** (A rotation matrix). The matrix

$$\mathbf{Q} = \begin{bmatrix} 0.6 & -0.8 \\ 0.8 & 0.6 \end{bmatrix}$$

rotates each vector in the plane counterclockwise by the angle  $\theta$  with  $\cos \theta = 0.6$  and  $\sin \theta = 0.8$  (about 53 degrees) (Banerjee and Roy 2014, chap. 8, Example 8.1, p. 207). It is orthogonal:

$$\begin{aligned}
 \mathbf{Q}^\top \mathbf{Q} &= \begin{bmatrix} 0.6 & 0.8 \\ -0.8 & 0.6 \end{bmatrix} \begin{bmatrix} 0.6 & -0.8 \\ 0.8 & 0.6 \end{bmatrix} && \text{(definition of the transpose)} \\
 &= \begin{bmatrix} 0.36 + 0.64 & -0.48 + 0.48 \\ -0.48 + 0.48 & 0.64 + 0.36 \end{bmatrix} && \text{(definition of matrix multiplication)} \\
 &= \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} && \text{(add)}
 \end{aligned}$$

*Remark 0.19* (The columns of an orthogonal matrix are orthonormal). Entry  $(i, j)$  of  $\mathbf{Q}^\top \mathbf{Q}$  is the dot product of column  $i$  and column  $j$  of  $\mathbf{Q}$ , so  $\mathbf{Q}^\top \mathbf{Q} = \mathbf{I}_p$  says that the columns of  $\mathbf{Q}$  are orthonormal (Definition 0.13). For a square matrix,  $\mathbf{Q}^\top \mathbf{Q} = \mathbf{I}_p$  also implies  $\mathbf{Q}\mathbf{Q}^\top = \mathbf{I}_p$ , so  $\mathbf{Q}^{-1} = \mathbf{Q}^\top$  (Definition 0.35) (Banerjee and Roy 2014, chap. 8, Theorem 8.1 and Definition 8.1, p. 209).

For example, the columns of  $\mathbf{Q}$  in Example 0.16 are  $(0.6, 0.8)$  and  $(-0.8, 0.6)$ . The diagonal

entries  $0.36 + 0.64 = 1$  of  $\mathbf{Q}^\top \mathbf{Q}$  are their squared norms, and the off-diagonal entries  $-0.48 + 0.48 = 0$  are their dot product. Multiplying in the other order also gives  $\mathbf{I}_2$ :

$$\begin{aligned}\mathbf{Q}\mathbf{Q}^\top &= \begin{bmatrix} 0.6 & -0.8 \\ 0.8 & 0.6 \end{bmatrix} \begin{bmatrix} 0.6 & 0.8 \\ -0.8 & 0.6 \end{bmatrix} && \text{(definition of the transpose)} \\ &= \begin{bmatrix} 0.36 + 0.64 & 0.48 - 0.48 \\ 0.48 - 0.48 & 0.64 + 0.36 \end{bmatrix} && \text{(definition of matrix multiplication)} \\ &= \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} && \text{(add)}\end{aligned}$$

So  $\mathbf{Q}^{-1} = \mathbf{Q}^\top$ .

**Theorem 0.17** (Orthogonal matrices preserve length). *If  $\mathbf{Q}$  is a  $p \times p$  orthogonal matrix (Definition 0.40) and  $\tilde{x}$  is a vector of length  $p$ , then*

$$\|\mathbf{Q}\tilde{x}\| = \|\tilde{x}\|$$

### **i** Proof

*Proof.*

$$\begin{aligned}\|\mathbf{Q}\tilde{x}\|^2 &= (\mathbf{Q}\tilde{x}) \cdot (\mathbf{Q}\tilde{x}) && \text{(definition of the Euclidean norm)} \\ &= (\mathbf{Q}\tilde{x})^\top (\mathbf{Q}\tilde{x}) && \text{(dot product as a matrix product)} \\ &= \tilde{x}^\top \mathbf{Q}^\top (\mathbf{Q}\tilde{x}) && \text{(transpose of a product)} \\ &= \tilde{x}^\top (\mathbf{Q}^\top \mathbf{Q}) \tilde{x} && \text{(regroup; matrix multiplication is associative)} \\ &= \tilde{x}^\top \mathbf{I}_p \tilde{x} && \text{(definition of an orthogonal matrix)} \\ &= \tilde{x}^\top \tilde{x} && \text{(identity matrix)} \\ &= \|\tilde{x}\|^2 && \text{(definition of the Euclidean norm)}\end{aligned}$$

Both norms are nonnegative square roots, so equal squares give  $\|\mathbf{Q}\tilde{x}\| = \|\tilde{x}\|$ . The second step is Example 0.7, the third is Theorem 0.10, and the fourth is Theorem 0.7.  $\square$

**Example 0.17** (Rotating a vector keeps its length). With  $\mathbf{Q}$  from Example 0.16 and  $\tilde{x} = (3, 4)$  from Example 0.4:

$$\begin{aligned}
\mathbf{Q}\tilde{x} &= \begin{bmatrix} 0.6 \cdot 3 - 0.8 \cdot 4 \\ 0.8 \cdot 3 + 0.6 \cdot 4 \end{bmatrix} && \text{(definition of matrix-vector multiplication)} \\
&= \begin{bmatrix} 1.8 - 3.2 \\ 2.4 + 2.4 \end{bmatrix} && \text{(multiply)} \\
&= \begin{bmatrix} -1.4 \\ 4.8 \end{bmatrix} && \text{(add)} \\
\|\mathbf{Q}\tilde{x}\| &= \sqrt{(-1.4)^2 + 4.8^2} && \text{(definition of the norm)} \\
&= \sqrt{1.96 + 23.04} && \text{(square)} \\
&= \sqrt{25} && \text{(add)} \\
&= 5 && \text{(take the square root)}
\end{aligned}$$

which is  $\|\tilde{x}\| = 5$ .

## Quadratic Forms

**Definition 0.41** (Quadratic form). A **quadratic form** is a mathematical expression of the structure

$$\tilde{x}^\top \mathbf{S} \tilde{x}$$

where  $\tilde{x}$  is a  $p \times 1$  vector and  $\mathbf{S}$  is a  $p \times p$  matrix.

*Remark 0.20* (Quadratic forms in statistics). A quadratic form extends the scalar expression  $cx^2$  to vectors: with  $p = 1$  and  $\mathbf{S} = [c]$ ,  $\tilde{x}^\top \mathbf{S} \tilde{x} = cx^2$ . With  $p = 2$ ,

$$\tilde{x}^\top \begin{bmatrix} 1 & 2 \\ 0 & 3 \end{bmatrix} \tilde{x} = x_1^2 + 2x_1x_2 + 3x_2^2,$$

which is  $1 + 4 + 12 = 17$  at  $\tilde{x} = (1, 2)$ .

Quadratic forms occur often in statistics:

- The residual sum of squares in linear regression (see [Vector Calculus](#)) is a quadratic form.
- The variance of a linear combination of estimates (see [Inference about Gaussian Linear Regression Models](#)) is a quadratic form:  $\text{Var}\left(\tilde{x}^\top \hat{\beta}\right) = \tilde{x}^\top \text{Var}\left(\hat{\beta}\right) \tilde{x}$ .

**Definition 0.42** (Symmetric part of a square matrix). The **symmetric part** of a  $p \times p$  matrix  $\mathbf{S}$  is

$$\frac{1}{2}(\mathbf{S} + \mathbf{S}^\top)$$

**Example 0.18** (The symmetric part of a  $2 \times 2$  matrix). For  $\mathbf{S} = \begin{bmatrix} 1 & 2 \\ 0 & 3 \end{bmatrix}$ , the symmetric part is

$$\frac{1}{2} \left( \begin{bmatrix} 1 & 2 \\ 0 & 3 \end{bmatrix} + \begin{bmatrix} 1 & 0 \\ 2 & 3 \end{bmatrix} \right) = \frac{1}{2} \begin{bmatrix} 2 & 2 \\ 2 & 6 \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ 1 & 3 \end{bmatrix},$$

which is symmetric.

**Theorem 0.18** (A quadratic form depends only on the symmetric part). *If  $\mathbf{S}$  is a  $p \times p$  matrix and  $\tilde{x}$  is a vector of length  $p$ , then*

$$\tilde{x}^\top \mathbf{S} \tilde{x} = \tilde{x}^\top \left( \frac{1}{2}(\mathbf{S} + \mathbf{S}^\top) \right) \tilde{x}.$$

*So the value of a quadratic form depends only on the symmetric part (Definition 0.42) of  $\mathbf{S}$ .*

### **i** Proof

*Proof.* The quadratic form  $\tilde{x}^\top \mathbf{S} \tilde{x}$  is a  $1 \times 1$  matrix, so it equals its own transpose:

$$\begin{aligned} \tilde{x}^\top \mathbf{S} \tilde{x} &= (\tilde{x}^\top \mathbf{S} \tilde{x})^\top && \text{(a } 1 \times 1 \text{ matrix is symmetric)} \\ &= \tilde{x}^\top \mathbf{S}^\top (\tilde{x}^\top)^\top && \text{(transpose of a product, twice)} \\ &= \tilde{x}^\top \mathbf{S}^\top \tilde{x} && \text{(transposing twice changes nothing)} \end{aligned}$$

Averaging the two expressions for  $\tilde{x}^\top \mathbf{S} \tilde{x}$ :

$$\begin{aligned} \tilde{x}^\top \mathbf{S} \tilde{x} &= \frac{1}{2}(\tilde{x}^\top \mathbf{S} \tilde{x} + \tilde{x}^\top \mathbf{S}^\top \tilde{x}) && \text{(average of two equal numbers)} \\ &= \frac{1}{2} \tilde{x}^\top (\mathbf{S} + \mathbf{S}^\top) \tilde{x} && \text{(distributive law)} \\ &= \tilde{x}^\top \left( \frac{1}{2}(\mathbf{S} + \mathbf{S}^\top) \right) \tilde{x} && \text{(move the scalar } \frac{1}{2} \text{ inside)} \end{aligned}$$

The distributive step is Theorem 0.8, and the transpose step is Theorem 0.10. □

**Example 0.19** (Replacing a matrix by its symmetric part). With  $\mathbf{S} = \begin{bmatrix} 1 & 2 \\ 0 & 3 \end{bmatrix}$  from Example 0.18 and  $\tilde{x} = (1, 1)$ :

$$\tilde{x}^\top \mathbf{S} \tilde{x} = 1 + 2 + 0 + 3 = 6 \quad \tilde{x}^\top \begin{bmatrix} 1 & 1 \\ 1 & 3 \end{bmatrix} \tilde{x} = 1 + 1 + 1 + 3 = 6$$

Both quadratic forms equal the sum of their matrix's entries, because every entry of  $\tilde{x}$  is 1.

## Trace and Matrix Inner Product

**Definition 0.43** (Trace). The **trace** of a  $p \times p$  matrix  $\mathbf{M}$  is the sum of its diagonal entries:

$$\text{tr}(\mathbf{M}) \stackrel{\text{def}}{=} \sum_{i=1}^p M_{ii}$$

**Example 0.20** (The trace of a  $2 \times 2$  matrix).

$$\text{tr}\left(\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}\right) = 1 + 4 = 5$$

**Theorem 0.19** (The trace of a product does not depend on the order). *For an  $m \times n$  matrix  $\mathbf{A}$  and an  $n \times m$  matrix  $\mathbf{B}$ :*

$$\text{tr}\left(\underbrace{\mathbf{AB}}_{m \times m}\right) = \text{tr}\left(\underbrace{\mathbf{BA}}_{n \times n}\right)$$

**i** Proof

*Proof.*

$$\begin{aligned}
 \text{tr}(\mathbf{AB}) &= \sum_{i=1}^m (\mathbf{AB})_{ii} && \text{(definition of the trace)} \\
 &= \sum_{i=1}^m \sum_{s=1}^n a_{is} b_{si} && \text{(definition of matrix multiplication)} \\
 &= \sum_{s=1}^n \sum_{i=1}^m a_{is} b_{si} && \text{(swap the order of two finite sums)} \\
 &= \sum_{s=1}^n \sum_{i=1}^m b_{si} a_{is} && \text{(multiplication of numbers is commutative)} \\
 &= \sum_{s=1}^n (\mathbf{BA})_{ss} && \text{(definition of matrix multiplication)} \\
 &= \text{tr}(\mathbf{BA}) && \text{(definition of the trace)}
 \end{aligned}$$

□

**Example 0.21** (Traces of the two products of a  $2 \times 3$  and a  $3 \times 2$  matrix). Let

$$\mathbf{A} = \begin{bmatrix} 1 & 2 & 0 \\ 0 & 1 & 3 \end{bmatrix} \quad \mathbf{B} = \begin{bmatrix} 1 & 0 \\ 2 & 1 \\ 0 & 1 \end{bmatrix}$$

The products are

$$\mathbf{AB} = \begin{bmatrix} 5 & 2 \\ 2 & 4 \end{bmatrix} \quad \mathbf{BA} = \begin{bmatrix} 1 & 2 & 0 \\ 2 & 5 & 3 \\ 0 & 1 & 3 \end{bmatrix}$$

The trace needs only the diagonal entries. For  $\mathbf{AB}$ , which is  $2 \times 2$ :

$$\begin{aligned}
 \text{tr}(\mathbf{AB}) &= (\mathbf{AB})_{11} + (\mathbf{AB})_{22} && \text{(definition of the trace)} \\
 &= (1 \cdot 1 + 2 \cdot 2 + 0 \cdot 0) + (0 \cdot 0 + 1 \cdot 1 + 3 \cdot 1) && \text{(definition of matrix multiplication)} \\
 &= (1 + 4 + 0) + (0 + 1 + 3) && \text{(multiply)} \\
 &= 5 + 4 && \text{(add within each entry)} \\
 &= 9 && \text{(add)}
 \end{aligned}$$

For  $\mathbf{BA}$ , which is  $3 \times 3$ :

$$\begin{aligned}
\text{tr}(\mathbf{BA}) &= (\mathbf{BA})_{11} + (\mathbf{BA})_{22} + (\mathbf{BA})_{33} && \text{(definition of the trace)} \\
&= (1 \cdot 1 + 0 \cdot 0) + (2 \cdot 2 + 1 \cdot 1) + (0 \cdot 0 + 1 \cdot 3) && \text{(definition of matrix multiplication)} \\
&= (1 + 0) + (4 + 1) + (0 + 3) && \text{(multiply)} \\
&= 1 + 5 + 3 && \text{(add within each entry)} \\
&= 9 && \text{(add)}
\end{aligned}$$

The products have different sizes, but their traces agree.

*Remark 0.21* (Moving a factor of a triple product). Let  $\mathbf{A}$  be  $m \times n$ ,  $\mathbf{B}$  be  $n \times k$ , and  $\mathbf{C}$  be  $k \times m$ , so that  $\mathbf{AB}$  is  $m \times k$  and  $\mathbf{ABC}$  is square. Applying Theorem 0.19 to the  $m \times k$  matrix  $\mathbf{AB}$  and the  $k \times m$  matrix  $\mathbf{C}$  moves the last factor of the triple product to the front (Banerjee and Roy 2014, chap. 1, Theorem 1.5 and eq. 1.17, p. 19):

$$\begin{aligned}
\text{tr}(\mathbf{ABC}) &= \text{tr}((\mathbf{AB})\mathbf{C}) && \text{(matrix multiplication is associative)} \\
&= \text{tr}(\mathbf{C}(\mathbf{AB})) && \text{(the theorem, applied to the two factors } \mathbf{AB} \text{ and } \mathbf{C}) \\
&= \text{tr}(\mathbf{CAB}) && \text{(matrix multiplication is associative)}
\end{aligned}$$

For example, let

$$\mathbf{A} = \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix} \quad \mathbf{B} = \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix} \quad \mathbf{C} = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$

Then

$$\mathbf{AB} = \begin{bmatrix} 1 \cdot 1 + 2 \cdot 1 & 1 \cdot 0 + 2 \cdot 1 \\ 0 \cdot 1 + 1 \cdot 1 & 0 \cdot 0 + 1 \cdot 1 \end{bmatrix} = \begin{bmatrix} 3 & 2 \\ 1 & 1 \end{bmatrix} \quad \mathbf{CA} = \begin{bmatrix} 0 \cdot 1 + 1 \cdot 0 & 0 \cdot 2 + 1 \cdot 1 \\ 1 \cdot 1 + 0 \cdot 0 & 1 \cdot 2 + 0 \cdot 1 \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ 1 & 2 \end{bmatrix}$$

so

$$(\mathbf{AB})\mathbf{C} = \begin{bmatrix} 3 \cdot 0 + 2 \cdot 1 & 3 \cdot 1 + 2 \cdot 0 \\ 1 \cdot 0 + 1 \cdot 1 & 1 \cdot 1 + 1 \cdot 0 \end{bmatrix} = \begin{bmatrix} 2 & 3 \\ 1 & 1 \end{bmatrix} \quad (\mathbf{CA})\mathbf{B} = \begin{bmatrix} 0 \cdot 1 + 1 \cdot 1 & 0 \cdot 0 + 1 \cdot 1 \\ 1 \cdot 1 + 2 \cdot 1 & 1 \cdot 0 + 2 \cdot 1 \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ 3 & 2 \end{bmatrix}$$

and both traces are the same:  $\text{tr}(\mathbf{ABC}) = 2 + 1 = 3$  and  $\text{tr}(\mathbf{CAB}) = 1 + 2 = 3$ .

**Definition 0.44** (Matrix inner product). The **inner product** of two  $n \times p$  matrices  $\mathbf{A}$  and  $\mathbf{B}$  is

$$\langle \mathbf{A}, \mathbf{B} \rangle \stackrel{\text{def}}{=} \text{tr} \left( \underbrace{\mathbf{A}^\top \mathbf{B}}_{p \times p} \right)$$

Banerjee and Roy (2014, chap. 15, eq. 15.7, p. 491) defines the same inner product on  $n \times p$  matrices with the trace.

**Theorem 0.20** (The matrix inner product multiplies matching entries). *For two  $n \times p$  matrices  $\mathbf{A}$  and  $\mathbf{B}$ :*

$$\langle \mathbf{A}, \mathbf{B} \rangle = \sum_{i=1}^n \sum_{j=1}^p a_{ij} b_{ij}$$

### **i** Proof

*Proof.*

$$\begin{aligned} \langle \mathbf{A}, \mathbf{B} \rangle &= \text{tr}(\mathbf{A}^\top \mathbf{B}) && \text{(definition of the inner product)} \\ &= \sum_{j=1}^p (\mathbf{A}^\top \mathbf{B})_{jj} && \text{(definition of the trace)} \\ &= \sum_{j=1}^p \sum_{i=1}^n (\mathbf{A}^\top)_{ji} b_{ij} && \text{(definition of matrix multiplication)} \\ &= \sum_{j=1}^p \sum_{i=1}^n a_{ij} b_{ij} && \text{(definition of the transpose)} \\ &= \sum_{i=1}^n \sum_{j=1}^p a_{ij} b_{ij} && \text{(swap the order of two finite sums)} \end{aligned}$$

□

**Example 0.22** (The inner product of two  $2 \times 2$  matrices). Let

$$\mathbf{A} = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \quad \mathbf{B} = \begin{bmatrix} 0 & 1 \\ -1 & 2 \end{bmatrix}$$

From Definition 0.44:

$$\begin{aligned}
\langle \mathbf{A}, \mathbf{B} \rangle &= \text{tr}(\mathbf{A}^\top \mathbf{B}) && \text{(definition of the inner product)} \\
&= \text{tr} \left( \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}^\top \begin{bmatrix} 0 & 1 \\ -1 & 2 \end{bmatrix} \right) && \text{(substitute)} \\
&= \text{tr} \left( \begin{bmatrix} 1 & 3 \\ 2 & 4 \end{bmatrix} \begin{bmatrix} 0 & 1 \\ -1 & 2 \end{bmatrix} \right) && \text{(definition of the transpose)} \\
&= \text{tr} \left( \begin{bmatrix} -3 & 7 \\ -4 & 10 \end{bmatrix} \right) && \text{(multiply)} \\
&= -3 + 10 && \text{(definition of the trace)} \\
&= 7 && \text{(add)}
\end{aligned}$$

From Theorem 0.20:

$$\begin{aligned}
\langle \mathbf{A}, \mathbf{B} \rangle &= 1 \cdot 0 + 2 \cdot 1 + 3 \cdot (-1) + 4 \cdot 2 && \text{(multiply matching entries and add)} \\
&= 0 + 2 - 3 + 8 && \text{(multiply)} \\
&= 7 && \text{(add)}
\end{aligned}$$

*Remark 0.22* (The matrix inner product as a dot product). Theorem 0.20 says that the matrix inner product is the dot product (Definition 0.5) of the two matrices' entries, each listed as one vector of length  $np$ , with both matrices' entries listed in the same order. For example, listing the entries of  $\mathbf{A}$  and  $\mathbf{B}$  in Example 0.22 row by row gives  $(1, 2, 3, 4)$  and  $(0, 1, -1, 2)$ , and  $(1, 2, 3, 4) \cdot (0, 1, -1, 2) = 7 = \langle \mathbf{A}, \mathbf{B} \rangle$ .

**Definition 0.45** (Frobenius norm). The **Frobenius norm** of an  $n \times p$  matrix  $\mathbf{A}$  is

$$\|\mathbf{A}\|_F \stackrel{\text{def}}{=} \sqrt{\langle \mathbf{A}, \mathbf{A} \rangle}$$

where  $\langle \cdot, \cdot \rangle$  is the matrix inner product (Definition 0.44).

**Example 0.23** (The Frobenius norm of a  $2 \times 2$  matrix). For  $\mathbf{A} = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$ :

$$\begin{aligned}
\|\mathbf{A}\|_F &= \sqrt{\langle \mathbf{A}, \mathbf{A} \rangle} && \text{(definition of the Frobenius norm)} \\
&= \sqrt{1 \cdot 1 + 2 \cdot 2 + 3 \cdot 3 + 4 \cdot 4} && \text{(sum of products of matching entries)} \\
&= \sqrt{1 + 4 + 9 + 16} && \text{(multiply)} \\
&= \sqrt{30} && \text{(add)}
\end{aligned}$$

The second step is Theorem 0.20.

*Remark 0.23* (The Frobenius norm is the Euclidean norm of the entries). By Theorem 0.20,  $\|\mathbf{A}\|_F^2 = \sum_{i=1}^n \sum_{j=1}^p a_{ij}^2$ , a sum of squares, so the square root is always defined.  $\|\mathbf{A}\|_F$  is the Euclidean norm (Definition 0.11) of the entries of  $\mathbf{A}$ , listed as one vector of length  $np$  (Banerjee and Roy 2014, chap. 15, Definition 15.4, p. 492).

For example, listing the entries of  $\mathbf{A}$  in Example 0.23 row by row gives the vector  $(1, 2, 3, 4)$  of length  $2 \cdot 2 = 4$ , and  $\|(1, 2, 3, 4)\| = \sqrt{1 + 4 + 9 + 16} = \sqrt{30} = \|\mathbf{A}\|_F$ .

## Matrix Decompositions

**Definition 0.46** (Eigenvalue and eigenvector). Let  $\mathbf{A}$  be a  $p \times p$  matrix. A real number  $\lambda$  is an **eigenvalue** of  $\mathbf{A}$  if some real vector  $\tilde{v} \neq \tilde{0}$  of length  $p$  satisfies

$$\underset{p \times p}{\mathbf{A}} \underset{p \times 1}{\tilde{v}} = \lambda \underset{p \times 1}{\tilde{v}}$$

Any such  $\tilde{v}$  is an **eigenvector** of  $\mathbf{A}$  for  $\lambda$ .

**Example 0.24** (Eigenvectors of a  $2 \times 2$  matrix). Let  $\mathbf{A} = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$ .

$(1, 1)$  is an eigenvector for the eigenvalue 3:

$$\begin{aligned} \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \end{bmatrix} &= \begin{bmatrix} 2+1 \\ 1+2 \end{bmatrix} && \text{(definition of matrix-vector multiplication)} \\ &= 3 \begin{bmatrix} 1 \\ 1 \end{bmatrix} && \text{(factor out 3)} \end{aligned}$$

$(1, -1)$  is an eigenvector for the eigenvalue 1:

$$\begin{aligned} \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix} \begin{bmatrix} 1 \\ -1 \end{bmatrix} &= \begin{bmatrix} 2-1 \\ 1-2 \end{bmatrix} && \text{(definition of matrix-vector multiplication)} \\ &= 1 \begin{bmatrix} 1 \\ -1 \end{bmatrix} && \text{(factor out 1)} \end{aligned}$$

*Remark 0.24* (Multiples of an eigenvector). Multiplying by  $\mathbf{A}$  rescales an eigenvector by  $\lambda$  and does not change the line it lies on. Any nonzero multiple  $c\tilde{v}$  of an eigenvector is also an eigenvector for the same  $\lambda$ :

$$\begin{aligned}
\mathbf{A}(c\tilde{v}) &= c\mathbf{A}\tilde{v} && \text{(move the scalar } c \text{ to the front)} \\
&= c\lambda\tilde{v} && (\tilde{v} \text{ is an eigenvector for } \lambda) \\
&= \lambda(c\tilde{v}) && \text{(multiplication of numbers is commutative)}
\end{aligned}$$

For example, in Example 0.24,  $2(1, 1) = (2, 2)$  is also an eigenvector for the eigenvalue 3:

$$\begin{aligned}
\begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix} \begin{bmatrix} 2 \\ 2 \end{bmatrix} &= \begin{bmatrix} 2 \cdot 2 + 1 \cdot 2 \\ 1 \cdot 2 + 2 \cdot 2 \end{bmatrix} && \text{(definition of matrix-vector multiplication)} \\
&= \begin{bmatrix} 4 + 2 \\ 2 + 4 \end{bmatrix} && \text{(multiply)} \\
&= \begin{bmatrix} 6 \\ 6 \end{bmatrix} && \text{(add)} \\
&= 3 \begin{bmatrix} 2 \\ 2 \end{bmatrix} && \text{(factor out 3)}
\end{aligned}$$

**Example 0.25** (A vector that is not an eigenvector). For the same  $\mathbf{A}$ ,  $(1, 0)$  is not an eigenvector:

$$\begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 2 \\ 1 \end{bmatrix},$$

and no number  $\lambda$  gives  $(2, 1) = \lambda(1, 0)$ , because the second entries would need  $1 = \lambda \cdot 0$ .

*Remark 0.25* (Real and complex eigenvalues). These notes take  $\lambda$  and  $\tilde{v}$  to be real. Banerjee and Roy (2014, chap. 11, Definition 11.1, pp. 312-313) also allows complex eigenvalues and eigenvectors, because some real matrices, such as a rotation by 90 degrees, have no real eigenvalues (Banerjee and Roy 2014, chap. 11, Example 11.2, p. 312).

For example,  $\mathbf{R} = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$  rotates each vector in the plane counterclockwise by 90 degrees:  $\mathbf{R}(v_1, v_2) = (-v_2, v_1)$ . Suppose  $\mathbf{R}\tilde{v} = \lambda\tilde{v}$  for a real number  $\lambda$ . Matching entries gives  $-v_2 = \lambda v_1$  and  $v_1 = \lambda v_2$ . Substituting the second equation into the first gives  $-v_2 = \lambda^2 v_2$ , so  $(1 + \lambda^2)v_2 = 0$ . Since  $1 + \lambda^2 > 0$ ,  $v_2 = 0$ , and then  $v_1 = \lambda v_2 = 0$ . So the only solution is  $\tilde{v} = \tilde{0}$ , and  $\mathbf{R}$  has no real eigenvalue.

**Theorem 0.21** (Spectral theorem for symmetric matrices). *If  $\mathbf{A}$  is a  $p \times p$  symmetric matrix (Definition 0.33) with real entries, then there are a  $p \times p$  orthogonal matrix  $\mathbf{Q}$  (Definition 0.40) and a  $p \times p$  diagonal matrix (Definition 0.34) with real diagonal entries  $\lambda_1, \dots, \lambda_p$  such that*

$$\underbrace{\mathbf{A}}_{p \times p} = \underbrace{\mathbf{Q}}_{p \times p} \underbrace{\underbrace{\mathbf{Q}^\top}_{p \times p}}_{p \times p}$$

Each  $\lambda_i$  is an eigenvalue of  $\mathbf{A}$  (Definition 0.46), and column  $i$  of  $\mathbf{Q}$  is an eigenvector of  $\mathbf{A}$  for  $\lambda_i$ .

*Remark 0.26* (An equivalent form of the spectral theorem). The proof, by induction on  $p$ , is outside the scope of these notes; see Banerjee and Roy (2014, chap. 11, Theorem 11.27, p. 349), which states the result in the equivalent form  $\mathbf{Q}^\top \mathbf{A} \mathbf{Q} = \mathbf{D}$ . The two forms are equivalent because  $\mathbf{Q}^\top \mathbf{Q} = \mathbf{Q} \mathbf{Q}^\top = \mathbf{I}_p$  (Remark 0.19). Multiplying  $\mathbf{A} = \mathbf{Q} \mathbf{Q}^\top \mathbf{D}$  by  $\mathbf{Q}^\top$  on the left and by  $\mathbf{Q}$  on the right gives

$$\begin{aligned} \mathbf{Q}^\top \mathbf{A} \mathbf{Q} &= \mathbf{Q}^\top (\mathbf{Q} \mathbf{Q}^\top) \mathbf{Q} && \text{(substitute)} \\ &= (\mathbf{Q}^\top \mathbf{Q}) (\mathbf{Q}^\top \mathbf{Q}) && \text{(regroup; matrix multiplication is associative)} \\ &= \mathbf{I}_p \mathbf{I}_p && (\mathbf{Q} \text{ is orthogonal}) \\ &= \mathbf{D} && \text{(multiplying by } \mathbf{I}_p \text{ changes nothing)} \end{aligned}$$

Multiplying  $\mathbf{Q}^\top \mathbf{A} \mathbf{Q} = \mathbf{D}$  by  $\mathbf{Q}$  on the left and by  $\mathbf{Q}^\top$  on the right gives back the first form:

$$\begin{aligned} \mathbf{Q} \mathbf{Q}^\top &= \mathbf{Q} (\mathbf{Q}^\top \mathbf{A} \mathbf{Q}) \mathbf{Q}^\top && \text{(substitute)} \\ &= (\mathbf{Q} \mathbf{Q}^\top) \mathbf{A} (\mathbf{Q} \mathbf{Q}^\top) && \text{(regroup; matrix multiplication is associative)} \\ &= \mathbf{I}_p \mathbf{A} \mathbf{I}_p && (\mathbf{Q} \mathbf{Q}^\top = \mathbf{I}_p) \\ &= \mathbf{A} && \text{(multiplying by } \mathbf{I}_p \text{ changes nothing)} \end{aligned}$$

For example, let  $\mathbf{A} = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$  and  $\mathbf{Q} = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$ , which is symmetric, so  $\mathbf{Q}^\top = \mathbf{Q}$ . Then

$$\begin{aligned}
\mathbf{Q}^\top \mathbf{A} \mathbf{Q} &= \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix} \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} && \text{(substitute)} \\
&= \frac{1}{\sqrt{2}} \cdot \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} && \text{(move the scalars to the front)} \\
&= \frac{1}{2} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} && \text{(multiply the scalars)} \\
&= \frac{1}{2} \begin{bmatrix} 1 \cdot 2 + 1 \cdot 1 & 1 \cdot 1 + 1 \cdot 2 \\ 1 \cdot 2 + (-1) \cdot 1 & 1 \cdot 1 + (-1) \cdot 2 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} && \text{(definition of matrix multiplication, first two matrices)} \\
&= \frac{1}{2} \begin{bmatrix} 2+1 & 1+2 \\ 2-1 & 1-2 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} && \text{(multiply)} \\
&= \frac{1}{2} \begin{bmatrix} 3 & 3 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} && \text{(add)} \\
&= \frac{1}{2} \begin{bmatrix} 3 \cdot 1 + 3 \cdot 1 & 3 \cdot 1 + 3 \cdot (-1) \\ 1 \cdot 1 + (-1) \cdot 1 & 1 \cdot 1 + (-1) \cdot (-1) \end{bmatrix} && \text{(definition of matrix multiplication)} \\
&= \frac{1}{2} \begin{bmatrix} 3+3 & 3-3 \\ 1-1 & 1+1 \end{bmatrix} && \text{(multiply)} \\
&= \frac{1}{2} \begin{bmatrix} 6 & 0 \\ 0 & 2 \end{bmatrix} && \text{(add)} \\
&= \begin{bmatrix} 3 & 0 \\ 0 & 1 \end{bmatrix} && \text{(multiply by } \frac{1}{2} \text{)}
\end{aligned}$$

a diagonal matrix with the eigenvalues 3 and 1 of  $\mathbf{A}$  from Example 0.24 on its diagonal.

**Definition 0.47** (Eigendecomposition). Let  $\mathbf{A}$  be a  $p \times p$  symmetric matrix with real entries. An **eigendecomposition**, or **spectral decomposition**, of  $\mathbf{A}$  is a factorization

$$\mathbf{A} = \mathbf{Q} \mathbf{Q}^\top$$

with  $\mathbf{Q}$  a  $p \times p$  orthogonal matrix (Definition 0.40) and a  $p \times p$  diagonal matrix (Definition 0.34). Theorem 0.21 says that every such  $\mathbf{A}$  has one.

**Example 0.26** (An eigendecomposition of a  $2 \times 2$  symmetric matrix). For  $\mathbf{A} = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$ , Example 0.24 found the eigenvectors  $(1, 1)$  for 3 and  $(1, -1)$  for 1. They are orthogonal (Definition 0.10):

$$\begin{aligned}
(1, 1) \cdot (1, -1) &= 1 \cdot 1 + 1 \cdot (-1) && \text{(definition of the dot product)} \\
&= 1 - 1 && \text{(multiply)} \\
&= 0 && \text{(add)}
\end{aligned}$$

Each has norm  $\sqrt{1^2 + 1^2} = \sqrt{2}$  (Definition 0.11), so dividing each by  $\sqrt{2}$  gives two orthonormal eigenvectors (Definition 0.13). Put them in the columns of  $\mathbf{Q}$ , and the matching eigenvalues on the diagonal of :

$$\mathbf{Q} = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} = \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \end{bmatrix}$$

This  $\mathbf{Q}$  is symmetric, so  $\mathbf{Q}^\top = \mathbf{Q}$ . Then

$$\begin{aligned}
\mathbf{Q}\mathbf{Q}^\top &= \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \end{bmatrix} && \text{(substitute)} \\
&= \frac{1}{\sqrt{2}} \cdot \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} && \text{(move the scalars to the front)} \\
&= \frac{1}{2} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} && \text{(multiply the scalars)} \\
&= \frac{1}{2} \begin{bmatrix} 1 \cdot 1 + 1 \cdot 1 & 1 \cdot 1 + 1 \cdot (-1) \\ 1 \cdot 1 + (-1) \cdot 1 & 1 \cdot 1 + (-1) \cdot (-1) \end{bmatrix} && \text{(definition of matrix multiplication, first two matrices)} \\
&= \frac{1}{2} \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix} && \text{(multiply)} \\
&= \frac{1}{2} \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix} && \text{(add)} \\
&= \frac{1}{2} \begin{bmatrix} 2 \cdot 1 + 0 \cdot 1 & 2 \cdot 1 + 0 \cdot (-1) \\ 0 \cdot 1 + 2 \cdot 1 & 0 \cdot 1 + 2 \cdot (-1) \end{bmatrix} && \text{(definition of matrix multiplication)} \\
&= \frac{1}{2} \begin{bmatrix} 2 & 2 \\ 2 & -2 \end{bmatrix} && \text{(multiply)} \\
&= \frac{1}{2} \begin{bmatrix} 2 & 2 \\ 2 & -2 \end{bmatrix} && \text{(add)} \\
&= \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} && \text{(multiply by } \frac{1}{2} \text{)}
\end{aligned}$$

which is  $\mathbf{A}$ .

*Remark 0.27* (An eigendecomposition is not unique). A symmetric matrix can have more than one eigendecomposition. Reordering the eigenvalues on the diagonal of , and the

columns of  $\mathbf{Q}$  with them, gives another one, and so does multiplying a column of  $\mathbf{Q}$  by  $-1$ .

For example, take  $\mathbf{A}$  from Example 0.26. Swapping the two eigenvalues and the two columns gives

$$\mathbf{Q}_1 = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ -1 & 1 \end{bmatrix} \quad \mathbf{\Lambda}_1 = \begin{bmatrix} 1 & 0 \\ 0 & 3 \end{bmatrix}$$

and multiplying the second column of  $\mathbf{Q}$  by  $-1$  gives

$$\mathbf{Q}_2 = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix} \quad \mathbf{\Lambda}_2 = \begin{bmatrix} 3 & 0 \\ 0 & 1 \end{bmatrix}$$

The columns of  $\mathbf{Q}_1$  and  $\mathbf{Q}_2$  are the columns of  $\mathbf{Q}$ , reordered or multiplied by  $-1$ , so they are still orthonormal, and  $\mathbf{Q}_1$  and  $\mathbf{Q}_2$  are orthogonal. Both products give back  $\mathbf{A}$ :

$$\begin{aligned} \mathbf{Q}_1 \mathbf{Q}_1^\top &= \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ -1 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & 3 \end{bmatrix} \left( \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ -1 & 1 \end{bmatrix} \right)^\top && \text{(substitute)} \\ &= \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ -1 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & 3 \end{bmatrix} \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix} && \text{(definition of the transpose)} \\ &= \frac{1}{\sqrt{2}} \cdot \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ -1 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & 3 \end{bmatrix} \begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix} && \text{(move the scalars to the front)} \\ &= \frac{1}{2} \begin{bmatrix} 1 & 1 \\ -1 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & 3 \end{bmatrix} \begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix} && \text{(multiply the scalars)} \\ &= \frac{1}{2} \begin{bmatrix} 1 \cdot 1 + 1 \cdot 0 & 1 \cdot 0 + 1 \cdot 3 \\ -1 \cdot 1 + 1 \cdot 0 & -1 \cdot 0 + 1 \cdot 3 \end{bmatrix} \begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix} && \text{(definition of matrix multiplication, first two matrices)} \\ &= \frac{1}{2} \begin{bmatrix} 1 + 0 & 0 + 3 \\ -1 + 0 & 0 + 3 \end{bmatrix} \begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix} && \text{(multiply)} \\ &= \frac{1}{2} \begin{bmatrix} 1 & 3 \\ -1 & 3 \end{bmatrix} \begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix} && \text{(add)} \\ &= \frac{1}{2} \begin{bmatrix} 1 \cdot 1 + 3 \cdot 1 & 1 \cdot (-1) + 3 \cdot 1 \\ -1 \cdot 1 + 3 \cdot 1 & -1 \cdot (-1) + 3 \cdot 1 \end{bmatrix} && \text{(definition of matrix multiplication)} \\ &= \frac{1}{2} \begin{bmatrix} 1 + 3 & -1 + 3 \\ -1 + 3 & 1 + 3 \end{bmatrix} && \text{(multiply)} \\ &= \frac{1}{2} \begin{bmatrix} 4 & 2 \\ 2 & 4 \end{bmatrix} && \text{(add)} \\ &= \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix} && \text{(multiply by } \frac{1}{2} \text{)} \end{aligned}$$

$$\begin{aligned}
\mathbf{Q}_{22}\mathbf{Q}_2^\top &= \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 3 & 0 \\ 0 & 1 \end{bmatrix} \left( \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix} \right)^\top && \text{(substitute)} \\
&= \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 3 & 0 \\ 0 & 1 \end{bmatrix} \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ -1 & 1 \end{bmatrix} && \text{(definition of the transpose)} \\
&= \frac{1}{\sqrt{2}} \cdot \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 3 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ -1 & 1 \end{bmatrix} && \text{(move the scalars to the front)} \\
&= \frac{1}{2} \begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 3 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ -1 & 1 \end{bmatrix} && \text{(multiply the scalars)} \\
&= \frac{1}{2} \begin{bmatrix} 1 \cdot 3 + (-1) \cdot 0 & 1 \cdot 0 + (-1) \cdot 1 \\ 1 \cdot 3 + 1 \cdot 0 & 1 \cdot 0 + 1 \cdot 1 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ -1 & 1 \end{bmatrix} && \text{(definition of matrix multiplication, first two matrices)} \\
&= \frac{1}{2} \begin{bmatrix} 3 + 0 & 0 - 1 \\ 3 + 0 & 0 + 1 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ -1 & 1 \end{bmatrix} && \text{(multiply)} \\
&= \frac{1}{2} \begin{bmatrix} 3 & -1 \\ 3 & 1 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ -1 & 1 \end{bmatrix} && \text{(add)} \\
&= \frac{1}{2} \begin{bmatrix} 3 \cdot 1 + (-1) \cdot (-1) & 3 \cdot 1 + (-1) \cdot 1 \\ 3 \cdot 1 + 1 \cdot (-1) & 3 \cdot 1 + 1 \cdot 1 \end{bmatrix} && \text{(definition of matrix multiplication)} \\
&= \frac{1}{2} \begin{bmatrix} 3 + 1 & 3 - 1 \\ 3 - 1 & 3 + 1 \end{bmatrix} && \text{(multiply)} \\
&= \frac{1}{2} \begin{bmatrix} 4 & 2 \\ 2 & 4 \end{bmatrix} && \text{(add)} \\
&= \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix} && \text{(multiply by } \frac{1}{2} \text{)}
\end{aligned}$$

**Theorem 0.22** (Singular value decomposition). *Let  $\mathbf{A}$  be an  $n \times p$  matrix with real entries and  $\text{rank}(\mathbf{A}) = r$  (Definition 0.26). Then there are an  $n \times n$  orthogonal matrix  $\mathbf{U}$ , a  $p \times p$  orthogonal matrix  $\mathbf{V}$  (Definition 0.40), and numbers  $\sigma_1 \geq \sigma_2 \geq \dots \geq \sigma_r > 0$  such that*

$$\underbrace{\mathbf{A}}_{n \times p} = \underbrace{\mathbf{U}}_{n \times n} \underbrace{\mathbf{D}}_{n \times p} \underbrace{\mathbf{V}^\top}_{p \times p}$$

where  $\mathbf{D}$  has entries  $d_{ii} = \sigma_i$  for  $i = 1, \dots, r$  and every other entry 0.

*Remark 0.28* (The SVD applies to every real matrix). The proof is outside the scope of these notes; see Banerjee and Roy (2014, chap. 12, Theorem 12.1, p. 373). Unlike the spectral theorem (Theorem 0.21), Theorem 0.22 applies to every real matrix, including

one that is not square or not symmetric.

For example, the  $1 \times 2$  matrix  $\mathbf{A} = \begin{bmatrix} 1 & 1 \end{bmatrix}$  is not square, so Theorem 0.21 does not apply to it. It has rank 1, and Theorem 0.22 holds with  $\mathbf{U} = [1]$ ,  $\mathbf{D} = \begin{bmatrix} \sqrt{2} & 0 \end{bmatrix}$ , and  $\mathbf{V}$  the orthogonal matrix  $\mathbf{Q}$  from Example 0.26, which is symmetric, so  $\mathbf{V}^\top = \mathbf{V}$ :

$$\begin{aligned}
 \mathbf{UDV}^\top &= [1] \begin{bmatrix} \sqrt{2} & 0 \end{bmatrix} \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} && \text{(substitute)} \\
 &= \begin{bmatrix} \sqrt{2} & 0 \end{bmatrix} \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} && \text{(multiplying by the } 1 \times 1 \text{ identity changes nothing)} \\
 &= \frac{1}{\sqrt{2}} \begin{bmatrix} \sqrt{2} & 0 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} && \text{(move the scalar to the front)} \\
 &= \frac{1}{\sqrt{2}} \begin{bmatrix} \sqrt{2} \cdot 1 + 0 \cdot 1 & \sqrt{2} \cdot 1 + 0 \cdot (-1) \end{bmatrix} && \text{(definition of matrix multiplication)} \\
 &= \frac{1}{\sqrt{2}} \begin{bmatrix} \sqrt{2} + 0 & \sqrt{2} + 0 \end{bmatrix} && \text{(multiply)} \\
 &= \frac{1}{\sqrt{2}} \begin{bmatrix} \sqrt{2} & \sqrt{2} \end{bmatrix} && \text{(add)} \\
 &= \begin{bmatrix} 1 & 1 \end{bmatrix} && \text{(divide by } \sqrt{2})
 \end{aligned}$$

$\mathbf{U}$  is orthogonal because  $\mathbf{U}^\top \mathbf{U} = [1] = \mathbf{I}_1$ , and the single singular value is  $\sigma_1 = \sqrt{2}$ .

**Definition 0.48** (Singular value decomposition and singular values). Let  $\mathbf{A}$  be an  $n \times p$  matrix with real entries and  $\text{rank}(\mathbf{A}) = r$ . A **singular value decomposition** (SVD) of  $\mathbf{A}$  is a factorization  $\mathbf{A} = \mathbf{UDV}^\top$  with  $\mathbf{U}$ ,  $\mathbf{D}$  and  $\mathbf{V}$  as in Theorem 0.22. The numbers  $\sigma_1 \geq \dots \geq \sigma_r > 0$  on the diagonal of  $\mathbf{D}$  are the **singular values** of  $\mathbf{A}$ .

**Example 0.27** (An SVD of a  $3 \times 2$  matrix). Let

$$\mathbf{A} = \begin{bmatrix} 1 & 1 \\ 1 & -1 \\ 1 & 1 \end{bmatrix} \quad \mathbf{U} = \begin{bmatrix} \frac{1}{\sqrt{2}} & 0 & \frac{1}{\sqrt{2}} \\ 0 & 1 & 0 \\ \frac{1}{\sqrt{2}} & 0 & -\frac{1}{\sqrt{2}} \end{bmatrix} \quad \mathbf{D} = \begin{bmatrix} 2 & 0 \\ 0 & \sqrt{2} \\ 0 & 0 \end{bmatrix} \quad \mathbf{V} = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$$

$\mathbf{V}$  is the orthogonal matrix  $\mathbf{Q}$  from Example 0.26.  $\mathbf{U}$  is orthogonal too. Entry  $(i, j)$  of  $\mathbf{U}^\top \mathbf{U}$  is the dot product of columns  $i$  and  $j$  of  $\mathbf{U}$ , and those columns are  $\tilde{u}_1 = (\frac{1}{\sqrt{2}}, 0, \frac{1}{\sqrt{2}})$ ,  $\tilde{u}_2 = (0, 1, 0)$  and  $\tilde{u}_3 = (\frac{1}{\sqrt{2}}, 0, -\frac{1}{\sqrt{2}})$ :

$$\begin{aligned}
\tilde{u}_1 \cdot \tilde{u}_1 &= \frac{1}{\sqrt{2}} \cdot \frac{1}{\sqrt{2}} + 0 \cdot 0 + \frac{1}{\sqrt{2}} \cdot \frac{1}{\sqrt{2}} && \text{(definition of the dot product)} \\
&= \frac{1}{2} + 0 + \frac{1}{2} && \text{(multiply)} \\
&= 1 && \text{(add)}
\end{aligned}$$

$$\begin{aligned}
\tilde{u}_2 \cdot \tilde{u}_2 &= 0 \cdot 0 + 1 \cdot 1 + 0 \cdot 0 && \text{(definition of the dot product)} \\
&= 0 + 1 + 0 && \text{(multiply)} \\
&= 1 && \text{(add)}
\end{aligned}$$

$$\begin{aligned}
\tilde{u}_3 \cdot \tilde{u}_3 &= \frac{1}{\sqrt{2}} \cdot \frac{1}{\sqrt{2}} + 0 \cdot 0 + \left(-\frac{1}{\sqrt{2}}\right) \cdot \left(-\frac{1}{\sqrt{2}}\right) && \text{(definition of the dot product)} \\
&= \frac{1}{2} + 0 + \frac{1}{2} && \text{(multiply)} \\
&= 1 && \text{(add)}
\end{aligned}$$

$$\begin{aligned}
\tilde{u}_1 \cdot \tilde{u}_2 &= \frac{1}{\sqrt{2}} \cdot 0 + 0 \cdot 1 + \frac{1}{\sqrt{2}} \cdot 0 && \text{(definition of the dot product)} \\
&= 0 + 0 + 0 && \text{(multiply)} \\
&= 0 && \text{(add)}
\end{aligned}$$

$$\begin{aligned}
\tilde{u}_1 \cdot \tilde{u}_3 &= \frac{1}{\sqrt{2}} \cdot \frac{1}{\sqrt{2}} + 0 \cdot 0 + \frac{1}{\sqrt{2}} \cdot \left(-\frac{1}{\sqrt{2}}\right) && \text{(definition of the dot product)} \\
&= \frac{1}{2} + 0 - \frac{1}{2} && \text{(multiply)} \\
&= 0 && \text{(add)}
\end{aligned}$$

$$\begin{aligned}
\tilde{u}_2 \cdot \tilde{u}_3 &= 0 \cdot \frac{1}{\sqrt{2}} + 1 \cdot 0 + 0 \cdot \left(-\frac{1}{\sqrt{2}}\right) && \text{(definition of the dot product)} \\
&= 0 + 0 + 0 && \text{(multiply)} \\
&= 0 && \text{(add)}
\end{aligned}$$

The dot product is symmetric, so these six values fill in all nine entries, and  $\mathbf{U}^\top \mathbf{U} = \mathbf{I}_3$ .  $\mathbf{V}$  is symmetric, so  $\mathbf{V}^\top = \mathbf{V}$ . Then

$$\begin{aligned}
\mathbf{D}\mathbf{V}^\top &= \begin{bmatrix} 2 & 0 \\ 0 & \sqrt{2} \\ 0 & 0 \end{bmatrix} \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} && \text{(substitute)} \\
&= \frac{1}{\sqrt{2}} \begin{bmatrix} 2 & 0 \\ 0 & \sqrt{2} \\ 0 & 0 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} && \text{(move the scalar to the front)} \\
&= \frac{1}{\sqrt{2}} \begin{bmatrix} 2 \cdot 1 + 0 \cdot 1 & 2 \cdot 1 + 0 \cdot (-1) \\ 0 \cdot 1 + \sqrt{2} \cdot 1 & 0 \cdot 1 + \sqrt{2} \cdot (-1) \\ 0 \cdot 1 + 0 \cdot 1 & 0 \cdot 1 + 0 \cdot (-1) \end{bmatrix} && \text{(definition of matrix multiplication)} \\
&= \frac{1}{\sqrt{2}} \begin{bmatrix} 2 + 0 & 2 + 0 \\ 0 + \sqrt{2} & 0 - \sqrt{2} \\ 0 + 0 & 0 + 0 \end{bmatrix} && \text{(multiply)} \\
&= \frac{1}{\sqrt{2}} \begin{bmatrix} 2 & 2 \\ \sqrt{2} & -\sqrt{2} \\ 0 & 0 \end{bmatrix} && \text{(add)} \\
&= \begin{bmatrix} \sqrt{2} & \sqrt{2} \\ 1 & -1 \\ 0 & 0 \end{bmatrix} && \text{(divide by } \sqrt{2})
\end{aligned}$$

and

$$\begin{aligned}
\mathbf{U}\mathbf{D}\mathbf{V}^\top &= \begin{bmatrix} \frac{1}{\sqrt{2}} & 0 & \frac{1}{\sqrt{2}} \\ 0 & 1 & 0 \\ \frac{1}{\sqrt{2}} & 0 & -\frac{1}{\sqrt{2}} \end{bmatrix} \begin{bmatrix} \sqrt{2} & \sqrt{2} \\ 1 & -1 \\ 0 & 0 \end{bmatrix} && \text{(substitute)} \\
&= \begin{bmatrix} \frac{1}{\sqrt{2}} \cdot \sqrt{2} + 0 \cdot 1 + \frac{1}{\sqrt{2}} \cdot 0 & \frac{1}{\sqrt{2}} \cdot \sqrt{2} + 0 \cdot (-1) + \frac{1}{\sqrt{2}} \cdot 0 \\ 0 \cdot \sqrt{2} + 1 \cdot 1 + 0 \cdot 0 & 0 \cdot \sqrt{2} + 1 \cdot (-1) + 0 \cdot 0 \\ \frac{1}{\sqrt{2}} \cdot \sqrt{2} + 0 \cdot 1 - \frac{1}{\sqrt{2}} \cdot 0 & \frac{1}{\sqrt{2}} \cdot \sqrt{2} + 0 \cdot (-1) - \frac{1}{\sqrt{2}} \cdot 0 \end{bmatrix} && \text{(definition of matrix multiplication)} \\
&= \begin{bmatrix} 1 + 0 + 0 & 1 + 0 + 0 \\ 0 + 1 + 0 & 0 - 1 + 0 \\ 1 + 0 - 0 & 1 + 0 - 0 \end{bmatrix} && \text{(multiply)} \\
&= \begin{bmatrix} 1 & 1 \\ 1 & -1 \\ 1 & 1 \end{bmatrix} && \text{(add)}
\end{aligned}$$

which is  $\mathbf{A}$ .  $\mathbf{A}$  has rank 2, because its two columns are linearly independent, and its singular values are  $\sigma_1 = 2$  and  $\sigma_2 = \sqrt{2}$ .

*Remark 0.29* (An SVD is not unique). An SVD is not unique (Banerjee and Roy 2014, chap. 12, Examples 12.2 and 12.3, p. 378). For any  $i \leq r$ , multiplying column  $i$  of both  $\mathbf{U}$  and  $\mathbf{V}$  by  $-1$  gives another one. The singular values do not depend on which SVD is chosen (Banerjee and Roy 2014, chap. 12, pp. 371 and 378).

For example, in Example 0.27, multiplying the first columns of  $\mathbf{U}$  and  $\mathbf{V}$  by  $-1$  gives

$$\mathbf{U}_1 = \begin{bmatrix} -\frac{1}{\sqrt{2}} & 0 & \frac{1}{\sqrt{2}} \\ 0 & 1 & 0 \\ -\frac{1}{\sqrt{2}} & 0 & -\frac{1}{\sqrt{2}} \end{bmatrix} \quad \mathbf{V}_1 = \frac{1}{\sqrt{2}} \begin{bmatrix} -1 & 1 \\ -1 & -1 \end{bmatrix}$$

Their columns are the columns of  $\mathbf{U}$  and  $\mathbf{V}$ , some multiplied by  $-1$ , so they are still orthonormal. With the same  $\mathbf{D}$ :

$$\begin{aligned} \mathbf{D}\mathbf{V}_1^\top &= \begin{bmatrix} 2 & 0 \\ 0 & \sqrt{2} \\ 0 & 0 \end{bmatrix} \left( \frac{1}{\sqrt{2}} \begin{bmatrix} -1 & 1 \\ -1 & -1 \end{bmatrix} \right)^\top && \text{(substitute)} \\ &= \begin{bmatrix} 2 & 0 \\ 0 & \sqrt{2} \\ 0 & 0 \end{bmatrix} \frac{1}{\sqrt{2}} \begin{bmatrix} -1 & -1 \\ 1 & -1 \end{bmatrix} && \text{(definition of the transpose)} \\ &= \frac{1}{\sqrt{2}} \begin{bmatrix} 2 & 0 \\ 0 & \sqrt{2} \\ 0 & 0 \end{bmatrix} \begin{bmatrix} -1 & -1 \\ 1 & -1 \end{bmatrix} && \text{(move the scalar to the front)} \\ &= \frac{1}{\sqrt{2}} \begin{bmatrix} 2 \cdot (-1) + 0 \cdot 1 & 2 \cdot (-1) + 0 \cdot (-1) \\ 0 \cdot (-1) + \sqrt{2} \cdot 1 & 0 \cdot (-1) + \sqrt{2} \cdot (-1) \\ 0 \cdot (-1) + 0 \cdot 1 & 0 \cdot (-1) + 0 \cdot (-1) \end{bmatrix} && \text{(definition of matrix multiplication)} \\ &= \frac{1}{\sqrt{2}} \begin{bmatrix} -2 + 0 & -2 + 0 \\ 0 + \sqrt{2} & 0 - \sqrt{2} \\ 0 + 0 & 0 + 0 \end{bmatrix} && \text{(multiply)} \\ &= \frac{1}{\sqrt{2}} \begin{bmatrix} -2 & -2 \\ \sqrt{2} & -\sqrt{2} \\ 0 & 0 \end{bmatrix} && \text{(add)} \\ &= \begin{bmatrix} -\sqrt{2} & -\sqrt{2} \\ 1 & -1 \\ 0 & 0 \end{bmatrix} && \text{(divide by } \sqrt{2}) \end{aligned}$$

and

$$\begin{aligned}
\mathbf{U}_1 \mathbf{D} \mathbf{V}_1^\top &= \begin{bmatrix} -\frac{1}{\sqrt{2}} & 0 & \frac{1}{\sqrt{2}} \\ 0 & 1 & 0 \\ -\frac{1}{\sqrt{2}} & 0 & -\frac{1}{\sqrt{2}} \end{bmatrix} \begin{bmatrix} -\sqrt{2} & -\sqrt{2} \\ 1 & -1 \\ 0 & 0 \end{bmatrix} && \text{(substitute)} \\
&= \begin{bmatrix} -\frac{1}{\sqrt{2}} \cdot (-\sqrt{2}) + 0 \cdot 1 + \frac{1}{\sqrt{2}} \cdot 0 & -\frac{1}{\sqrt{2}} \cdot (-\sqrt{2}) + 0 \cdot (-1) + \frac{1}{\sqrt{2}} \cdot 0 \\ 0 \cdot (-\sqrt{2}) + 1 \cdot 1 + 0 \cdot 0 & 0 \cdot (-\sqrt{2}) + 1 \cdot (-1) + 0 \cdot 0 \\ -\frac{1}{\sqrt{2}} \cdot (-\sqrt{2}) + 0 \cdot 1 - \frac{1}{\sqrt{2}} \cdot 0 & -\frac{1}{\sqrt{2}} \cdot (-\sqrt{2}) + 0 \cdot (-1) - \frac{1}{\sqrt{2}} \cdot 0 \end{bmatrix} && \text{(definition of matrix multiplication)} \\
&= \begin{bmatrix} 1 + 0 + 0 & 1 + 0 + 0 \\ 0 + 1 + 0 & 0 - 1 + 0 \\ 1 + 0 - 0 & 1 + 0 - 0 \end{bmatrix} && \text{(multiply)} \\
&= \begin{bmatrix} 1 & 1 \\ 1 & -1 \\ 1 & 1 \end{bmatrix} && \text{(add)}
\end{aligned}$$

which is  $\mathbf{A}$  again. Both SVDs have the singular values  $\sigma_1 = 2$  and  $\sigma_2 = \sqrt{2}$ .

*Remark 0.30* (Counting zero singular values). Some texts, and R's `svd()`, also count  $\min(n, p) - r$  singular values equal to 0, so that every  $n \times p$  matrix has  $\min(n, p)$  singular values.

For example,  $\mathbf{B} = \begin{bmatrix} 3 & 0 \\ 0 & 0 \end{bmatrix}$  has rank  $r = 1$ . Taking  $\mathbf{U} = \mathbf{V} = \mathbf{I}_2$  and  $\mathbf{D} = \mathbf{B}$  gives an SVD  $\mathbf{B} = \mathbf{I}_2 \mathbf{B} \mathbf{I}_2^\top$ , so by Definition 0.48,  $\mathbf{B}$  has one singular value,  $\sigma_1 = 3$ . R's `svd()` reports  $\min(2, 2) = 2$  singular values for  $\mathbf{B}$ : 3 and 0.

**Theorem 0.23** (An SVD gives an eigendecomposition of  $\mathbf{A}^\top \mathbf{A}$ ). *Let  $\mathbf{A} = \mathbf{U} \mathbf{D} \mathbf{V}^\top$  be a singular value decomposition (Definition 0.48) of an  $n \times p$  matrix  $\mathbf{A}$  with singular values  $\sigma_1, \dots, \sigma_r$ . Then  $\mathbf{A}^\top \mathbf{A}$  is symmetric (Definition 0.33), and*

$$\underbrace{\mathbf{A}^\top \mathbf{A}}_{p \times p} = \underbrace{\mathbf{V}}_{p \times p} \underbrace{\mathbf{D}^\top \mathbf{D}}_{p \times p} \underbrace{\mathbf{V}^\top}_{p \times p}$$

where  $\stackrel{\text{def}}{=} \mathbf{D}^\top \mathbf{D}$  is the  $p \times p$  diagonal matrix whose  $i$ -th diagonal entry  $\lambda_i$  is  $\sigma_i^2$  for  $i \leq r$  and 0 for  $i > r$ . So  $\mathbf{V} \mathbf{V}^\top$  is an eigendecomposition (Definition 0.47) of  $\mathbf{A}^\top \mathbf{A}$ : column  $i$  of  $\mathbf{V}$  is an eigenvector of  $\mathbf{A}^\top \mathbf{A}$  for the eigenvalue  $\lambda_i$ .

## i Proof

*Proof. Symmetry.*

$$\begin{aligned} (\mathbf{A}^\top \mathbf{A})^\top &= \mathbf{A}^\top (\mathbf{A}^\top)^\top && \text{(transpose of a product)} \\ &= \mathbf{A}^\top \mathbf{A} && \text{(transposing twice changes nothing)} \end{aligned}$$

The first step is Theorem 0.10, and the second follows from Definition 0.16, which swaps rows and columns, so swapping them again restores  $\mathbf{A}$ .

**The factorization.**

$$\begin{aligned} \mathbf{A}^\top \mathbf{A} &= (\mathbf{U} \mathbf{D} \mathbf{V}^\top)^\top \mathbf{U} \mathbf{D} \mathbf{V}^\top && \text{(substitute the SVD)} \\ &= (\mathbf{D} \mathbf{V}^\top)^\top \mathbf{U}^\top \mathbf{U} \mathbf{D} \mathbf{V}^\top && \text{(transpose of the product of } \mathbf{U} \text{ and } \mathbf{D} \mathbf{V}^\top) \\ &= (\mathbf{V}^\top)^\top \mathbf{D}^\top \mathbf{U}^\top \mathbf{U} \mathbf{D} \mathbf{V}^\top && \text{(transpose of the product of } \mathbf{D} \text{ and } \mathbf{V}^\top) \\ &= \mathbf{V} \mathbf{D}^\top \mathbf{U}^\top \mathbf{U} \mathbf{D} \mathbf{V}^\top && \text{(transposing twice changes nothing)} \\ &= \mathbf{V} \mathbf{D}^\top (\mathbf{U}^\top \mathbf{U}) \mathbf{D} \mathbf{V}^\top && \text{(regroup; matrix multiplication is associative)} \\ &= \mathbf{V} \mathbf{D}^\top \mathbf{I}_n \mathbf{D} \mathbf{V}^\top && \text{(} \mathbf{U} \text{ is orthogonal)} \\ &= \mathbf{V} (\mathbf{D}^\top \mathbf{D}) \mathbf{V}^\top && \text{(identity matrix)} \\ &= \mathbf{V} \mathbf{V}^\top && \text{(definition of )} \end{aligned}$$

**The entries of  $\lambda$ .** Write  $\lambda_{ij}$  for entry  $(i, j)$  of  $\lambda$ , so  $\lambda_i = \lambda_{ii}$ . Only the entries  $d_{kk}$  with  $k \leq r$  of  $\mathbf{D}$  can be nonzero.

$$\begin{aligned} \lambda_{ij} &= \sum_{k=1}^n (\mathbf{D}^\top)_{ik} d_{kj} && \text{(definition of matrix multiplication)} \\ &= \sum_{k=1}^n d_{ki} d_{kj} && \text{(definition of the transpose)} \end{aligned}$$

The term for  $k$  is nonzero only if  $k = i \leq r$  and  $k = j$ . So  $\lambda_{ij} = 0$  when  $i \neq j$ . On the diagonal, only the term  $k = i$  can be nonzero, and for  $i \leq r$

$$\begin{aligned} \lambda_i &= \lambda_{ii} && \text{(notation)} \\ &= d_{ii} d_{ii} && \text{(the only term that can be nonzero)} \\ &= \sigma_i \cdot \sigma_i && \text{(definition of } \mathbf{D}) \\ &= \sigma_i^2 && \text{(notation for a square)} \end{aligned}$$

For  $i > r$ , every term is 0, so  $\lambda_i = 0$ .

**The eigenvectors.** Write  $\tilde{v}_i$  for column  $i$  of  $\mathbf{V}$ , and  $\tilde{e}_i$  for the indicator vector (Definition 0.9) of length  $p$ . Column  $i$  of  $\mathbf{V}^\top \mathbf{V} = \mathbf{I}_p$  says  $\mathbf{V}^\top \tilde{v}_i = \tilde{e}_i$ . Then

$$\begin{aligned}
\mathbf{A}^\top \mathbf{A} \tilde{v}_i &= \mathbf{V} \mathbf{V}^\top \tilde{v}_i && \text{(the factorization)} \\
&= \mathbf{V} \tilde{e}_i && (\mathbf{V}^\top \tilde{v}_i = \tilde{e}_i) \\
&= \mathbf{V} (\lambda_i \tilde{e}_i) && \text{(column } i \text{ of the diagonal matrix )} \\
&= \lambda_i \mathbf{V} \tilde{e}_i && \text{(move the scalar } \lambda_i \text{ to the front)} \\
&= \lambda_i \tilde{v}_i && (\mathbf{V} \tilde{e}_i \text{ is column } i \text{ of } \mathbf{V})
\end{aligned}$$

and  $\tilde{v}_i \neq \tilde{0}$  because it has norm 1. □

**Example 0.28** ( $\mathbf{A}^\top \mathbf{A}$  for the matrix of the SVD example). For  $\mathbf{A}$  in Example 0.27:

$$\begin{aligned}
\mathbf{A}^\top \mathbf{A} &= \begin{bmatrix} 1 & 1 & 1 \\ 1 & -1 & 1 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 1 & -1 \\ 1 & 1 \end{bmatrix} && \text{(definition of the transpose)} \\
&= \begin{bmatrix} 1+1+1 & 1-1+1 \\ 1-1+1 & 1+1+1 \end{bmatrix} && \text{(definition of matrix multiplication)} \\
&= \begin{bmatrix} 3 & 1 \\ 1 & 3 \end{bmatrix} && \text{(add)}
\end{aligned}$$

Theorem 0.23 says its eigenvalues are  $\sigma_1^2 = 4$  and  $\sigma_2^2 = 2$ , with the columns of  $\mathbf{V}$  as eigenvectors. Checking the first column, up to its factor  $\frac{1}{\sqrt{2}}$ :

$$\begin{aligned}
\begin{bmatrix} 3 & 1 \\ 1 & 3 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \end{bmatrix} &= \begin{bmatrix} 3 \cdot 1 + 1 \cdot 1 \\ 1 \cdot 1 + 3 \cdot 1 \end{bmatrix} && \text{(definition of matrix-vector multiplication)} \\
&= \begin{bmatrix} 3+1 \\ 1+3 \end{bmatrix} && \text{(multiply)} \\
&= \begin{bmatrix} 4 \\ 4 \end{bmatrix} && \text{(add)} \\
&= 4 \begin{bmatrix} 1 \\ 1 \end{bmatrix} && \text{(factor out 4)}
\end{aligned}$$

and the second:

$$\begin{aligned}
\begin{bmatrix} 3 & 1 \\ 1 & 3 \end{bmatrix} \begin{bmatrix} 1 \\ -1 \end{bmatrix} &= \begin{bmatrix} 3 \cdot 1 + 1 \cdot (-1) \\ 1 \cdot 1 + 3 \cdot (-1) \end{bmatrix} && \text{(definition of matrix-vector multiplication)} \\
&= \begin{bmatrix} 3-1 \\ 1-3 \end{bmatrix} && \text{(multiply)} \\
&= \begin{bmatrix} 2 \\ -2 \end{bmatrix} && \text{(add)} \\
&= 2 \begin{bmatrix} 1 \\ -1 \end{bmatrix} && \text{(factor out 2)}
\end{aligned}$$

*Remark 0.31* (Building an SVD from an eigendecomposition). Theorem 0.23 matches Banerjee and Roy (2014, chap. 12, pp. 372-373), which works in the other direction: it constructs an SVD from a spectral decomposition of  $\mathbf{A}^\top \mathbf{A}$  and sets  $\sigma_i = \sqrt{\lambda_i}$ . For example, in Example 0.28 the eigenvalues of  $\mathbf{A}^\top \mathbf{A}$  are 4 and 2, and  $\sqrt{4} = 2$  and  $\sqrt{2}$  are the singular values  $\sigma_1$  and  $\sigma_2$  of  $\mathbf{A}$  from Example 0.27.

## Definite Matrices

**Definition 0.49** (Positive semidefinite matrix). A  $p \times p$  matrix  $\mathbf{A}$  is **positive semidefinite** if it satisfies both conditions:

- $\mathbf{A}$  is symmetric (Definition 0.33).
- Every quadratic form (Definition 0.41) in  $\mathbf{A}$  is non-negative:  $\tilde{x}^\top \mathbf{A} \tilde{x} \geq 0$  for every vector  $\tilde{x}$  of length  $p$ .

**Example 0.29** (A positive semidefinite matrix). Let  $\mathbf{B} = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$ , which is symmetric. For any  $\tilde{x} = (x_1, x_2)$ :

$$\begin{aligned} \tilde{x}^\top \mathbf{B} \tilde{x} &= x_1^2 + x_1 x_2 + x_2 x_1 + x_2^2 && \text{(multiply out the quadratic form)} \\ &= (x_1 + x_2)^2 && \text{(complete the square)} \\ &\geq 0 && \text{(a square is non-negative)} \end{aligned}$$

So  $\mathbf{B}$  is positive semidefinite.

**Definition 0.50** (Positive definite matrix). A  $p \times p$  matrix  $\mathbf{A}$  is **positive definite** if it satisfies both conditions:

- $\mathbf{A}$  is symmetric (Definition 0.33).
- Every quadratic form (Definition 0.41) in  $\mathbf{A}$  at a nonzero vector is positive:  $\tilde{x}^\top \mathbf{A} \tilde{x} > 0$  for every vector  $\tilde{x} \neq \tilde{0}$  of length  $p$ .

**Example 0.30** (Positive definite, semidefinite, and neither).

- The identity matrix  $\mathbf{I}_p$  (Definition 0.32) is positive definite:  $\tilde{x}^\top \mathbf{I}_p \tilde{x} = \sum_{i=1}^p x_i^2$ , which is positive unless every  $x_i$  is 0.
- $\mathbf{B} = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$  from Example 0.29 is positive semidefinite but not positive definite: at  $\tilde{x} = (1, -1) \neq \tilde{0}$ ,  $\tilde{x}^\top \mathbf{B} \tilde{x} = (1 - 1)^2 = 0$ .

- $\mathbf{D} = \begin{bmatrix} 1 & 2 \\ 2 & 1 \end{bmatrix}$  is symmetric but not positive semidefinite: at  $\tilde{x} = (1, -1)$ ,  $\tilde{x}^\top \mathbf{D} \tilde{x} = 1 - 2 - 2 + 1 = -2 < 0$ .

*Remark 0.32* (Why the definition requires symmetry). A positive definite matrix is positive semidefinite (Definition 0.49):  $\tilde{x}^\top \mathbf{A} \tilde{x} > 0$  for every  $\tilde{x} \neq \tilde{0}$ , and  $\tilde{0}^\top \mathbf{A} \tilde{0} = 0$ . For example,  $\mathbf{I}_p$  in Example 0.30 is both.

Some sources drop the symmetry condition from both definitions. These notes keep it, because without it a matrix can pass the quadratic-form condition and still have no real eigenvalues (Definition 0.46). For example,  $\mathbf{C} = \begin{bmatrix} 1 & 1 \\ -1 & 1 \end{bmatrix}$  is not symmetric, and for any  $\tilde{x} = (x_1, x_2)$ :

$$\begin{aligned} \tilde{x}^\top \mathbf{C} \tilde{x} &= x_1(x_1 + x_2) + x_2(-x_1 + x_2) && \text{(multiply out the quadratic form)} \\ &= x_1^2 + x_1x_2 - x_2x_1 + x_2^2 && \text{(distribute)} \\ &= x_1^2 + x_2^2 && \text{(the middle terms cancel)} \end{aligned}$$

which is positive for every  $\tilde{x} \neq \tilde{0}$ . But suppose  $\mathbf{C}\tilde{v} = \lambda\tilde{v}$  for a real number  $\lambda$ . Matching entries gives  $v_1 + v_2 = \lambda v_1$  and  $-v_1 + v_2 = \lambda v_2$ , so  $v_2 = (\lambda - 1)v_1$  and  $-v_1 = (\lambda - 1)v_2$ . Substituting the first into the second gives  $-v_1 = (\lambda - 1)^2 v_1$ , so  $(1 + (\lambda - 1)^2)v_1 = 0$ . Since  $1 + (\lambda - 1)^2 > 0$ ,  $v_1 = 0$ , and then  $v_2 = (\lambda - 1)v_1 = 0$ . So no real  $\lambda$  has an eigenvector  $\tilde{v} \neq \tilde{0}$ .

See also [https://en.wikipedia.org/wiki/Definite\\_matrix](https://en.wikipedia.org/wiki/Definite_matrix).

**Theorem 0.24** (Definiteness and eigenvalues). *Let  $\mathbf{A}$  be a  $p \times p$  symmetric matrix with real entries, with eigendecomposition  $\mathbf{A} = \mathbf{Q}\mathbf{Q}^\top$  (Definition 0.47) and eigenvalues  $\lambda_1, \dots, \lambda_p$  on the diagonal of  $\mathbf{\Lambda}$ . Then:*

- $\mathbf{A}$  is positive semidefinite (Definition 0.49) if and only if every  $\lambda_i \geq 0$ .
- $\mathbf{A}$  is positive definite (Definition 0.50) if and only if every  $\lambda_i > 0$ .

#### **i** Proof

*Proof.* For any vector  $\tilde{x}$  of length  $p$ , let  $\tilde{y} = \mathbf{Q}^\top \tilde{x}$ . Then:

$$\begin{aligned}
\tilde{x}^\top \mathbf{A} \tilde{x} &= \tilde{x}^\top \mathbf{Q} \mathbf{Q}^\top \tilde{x} && \text{(substitute the eigendecomposition)} \\
&= (\mathbf{Q}^\top \tilde{x})^\top (\mathbf{Q}^\top \tilde{x}) && \text{(transpose of a product)} \\
&= \tilde{y}^\top \tilde{y} && \text{(definition of } \tilde{y}) \\
&= \sum_{i=1}^p \lambda_i y_i^2 && \text{(is diagonal)}
\end{aligned}$$

The second step is Theorem 0.10. Also,  $\tilde{x} = \tilde{0}$  exactly when  $\tilde{y} = \tilde{0}$ :  $\mathbf{Q} \mathbf{Q}^\top = \mathbf{I}_p$  for an orthogonal matrix (Definition 0.40), so  $\tilde{x} = \mathbf{Q} \tilde{y}$ .

If every  $\lambda_i \geq 0$ , then every term  $\lambda_i y_i^2 \geq 0$ , so  $\tilde{x}^\top \mathbf{A} \tilde{x} \geq 0$ . If every  $\lambda_i > 0$  and  $\tilde{x} \neq \tilde{0}$ , then some  $y_i \neq 0$ , so at least one term is positive and the rest are non-negative, and  $\tilde{x}^\top \mathbf{A} \tilde{x} > 0$ . Conversely, take  $\tilde{x} = \tilde{q}_i$ , column  $i$  of  $\mathbf{Q}$ , which is not  $\tilde{0}$  because it has norm 1. Then  $\tilde{y} = \mathbf{Q}^\top \tilde{q}_i$  is column  $i$  of  $\mathbf{Q}^\top \mathbf{Q} = \mathbf{I}_p$ , so  $y_i = 1$  and every other entry of  $\tilde{y}$  is 0, and the display gives  $\tilde{q}_i^\top \mathbf{A} \tilde{q}_i = \lambda_i$ . So if  $\mathbf{A}$  is positive semidefinite,  $\lambda_i \geq 0$ , and if  $\mathbf{A}$  is positive definite,  $\lambda_i > 0$ .  $\square$

**Example 0.31** (Reading definiteness off the eigenvalues).

- $\mathbf{A} = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$  has eigenvalues 3 and 1 (Example 0.24), both positive, so it is positive definite.
- $\mathbf{B} = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$  has eigenvalues 2, for the eigenvector  $(1, 1)$ , and 0, for the eigenvector  $(1, -1)$ , so it is positive semidefinite but not positive definite, as Example 0.30 found directly.

**Theorem 0.25** (A positive definite matrix has a positive definite inverse). *Let  $\mathbf{A}$  be a  $p \times p$  positive definite matrix (Definition 0.50) with real entries, with eigendecomposition  $\mathbf{A} = \mathbf{Q} \mathbf{Q}^\top$  (Definition 0.47) and eigenvalues  $\lambda_1, \dots, \lambda_p$ . Then  $\mathbf{A}$  is invertible (Definition 0.36),*

$$\mathbf{A}^{-1} = \mathbf{Q}^{-1} \mathbf{Q}^\top, \quad {}^{-1} = \begin{bmatrix} 1/\lambda_1 & \cdots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \cdots & 1/\lambda_p \end{bmatrix},$$

and  $\mathbf{A}^{-1}$  is positive definite.

### **i** Proof

*Proof.* By Theorem 0.24, every  $\lambda_i > 0$ , so  ${}^{-1}$  is well defined, and multiplying the two diagonal matrices entry by entry gives  ${}^{-1} = {}^{-1} = \mathbf{I}_p$ . Write  $\mathbf{B} = \mathbf{Q}^{-1} \mathbf{Q}^\top$ . Then:

$$\begin{aligned}
\mathbf{AB} &= \mathbf{QQ}^\top \mathbf{Q}^{-1} \mathbf{Q}^\top && \text{(substitute)} \\
&= \mathbf{Q} \mathbf{I}_p^{-1} \mathbf{Q}^\top && (\mathbf{Q}^\top \mathbf{Q} = \mathbf{I}_p) \\
&= \mathbf{Q}^{-1} \mathbf{Q}^\top && \text{(identity matrix)} \\
&= \mathbf{QQ}^\top && (-1 = \mathbf{I}_p) \\
&= \mathbf{I}_p && (\mathbf{Q} \text{ is orthogonal})
\end{aligned}$$

The same steps, with  $^{-1}$  and  $^\top$  swapped, give  $\mathbf{BA} = \mathbf{I}_p$ , so  $\mathbf{B} = \mathbf{A}^{-1}$  (Definition 0.35). The steps with  $\mathbf{Q}$  use Definition 0.40, whose remark records that  $\mathbf{QQ}^\top = \mathbf{I}_p$  too, and the identity step uses Theorem 0.12.

$\mathbf{A}^{-1}$  is symmetric by Corollary 0.1. For  $\tilde{x} \neq \tilde{0}$ , let  $\tilde{y} = \mathbf{Q}^\top \tilde{x}$ , which is not  $\tilde{0}$  because  $\tilde{x} = \mathbf{Q}\tilde{y}$ . As in the proof of Theorem 0.24:

$$\begin{aligned}
\tilde{x}^\top \mathbf{A}^{-1} \tilde{x} &= \tilde{y}^\top {}^{-1} \tilde{y} && \text{(substitute } \mathbf{A}^{-1} = \mathbf{Q}^{-1} \mathbf{Q}^\top \text{)} \\
&= \sum_{i=1}^p \frac{y_i^2}{\lambda_i} && (-1 \text{ is diagonal)} \\
&> 0 && \text{(each } \lambda_i > 0, \text{ and some } y_i \neq 0)
\end{aligned}$$

So  $\mathbf{A}^{-1}$  is positive definite. □

**Example 0.32** (Inverting a positive definite matrix). For  $\mathbf{A} = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$ , Example 0.26

gives  $\mathbf{Q} = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$  and eigenvalues 3 and 1, so:

$$\begin{aligned}
\mathbf{A}^{-1} &= \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} 1/3 & 0 \\ 0 & 1 \end{bmatrix} \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} && \text{(substitute; } \mathbf{Q} \text{ is symmetric)} \\
&= \frac{1}{2} \begin{bmatrix} 1/3 & 1 \\ 1/3 & -1 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} && \text{(multiply the first two matrices)} \\
&= \frac{1}{2} \begin{bmatrix} 4/3 & -2/3 \\ -2/3 & 4/3 \end{bmatrix} && \text{(multiply)} \\
&= \frac{1}{3} \begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix} && \text{(simplify)}
\end{aligned}$$

Multiplying out,  $\begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix} \cdot \frac{1}{3} \begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix} = \frac{1}{3} \begin{bmatrix} 3 & 0 \\ 0 & 3 \end{bmatrix} = \mathbf{I}_2$ .

## Determinants

**Definition 0.51** (Determinant). The **determinant** of a  $p \times p$  matrix  $\mathbf{A}$  with entries  $a_{ij}$ , written  $\det(\mathbf{A})$  or  $|\mathbf{A}|$ , is the number defined recursively in  $p$ :

- For  $p = 1$ ,  $\det(\mathbf{A}) = a_{11}$ .
- For  $p \geq 2$ ,

$$\det(\mathbf{A}) = \sum_{j=1}^p (-1)^{1+j} a_{1j} \det(\mathbf{A}_{(1j)}),$$

where  $\mathbf{A}_{(1j)}$  is the  $(p-1) \times (p-1)$  matrix left after deleting row 1 and column  $j$  of  $\mathbf{A}$ .

**Example 0.33** (Determinant of a  $2 \times 2$  matrix). For  $\mathbf{A} = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$ , deleting row 1 and column 1 leaves  $[d]$ , and deleting row 1 and column 2 leaves  $[c]$ . So:

$$\begin{aligned} \det(\mathbf{A}) &= (-1)^{1+1} a \det([d]) + (-1)^{1+2} b \det([c]) && \text{(definition, } p = 2) \\ &= ad - bc && \text{(definition, } p = 1) \end{aligned}$$

For example,  $\det\left(\begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}\right) = 2 \cdot 2 - 1 \cdot 1 = 3$ .

*Remark 0.33* (Cofactor expansion). The recursive formula in Definition 0.51 is the *cofactor expansion* along the first row. Other sources define the determinant as a sum over all orderings of the columns and derive this expansion from it; the two definitions agree (Banerjee and Roy 2014).

Each ordering picks one entry from each row, from the column that the ordering assigns to that row, and multiplies them. Its sign is  $+1$  if the ordering takes an even number of swaps of two columns to reach from  $(1, 2, \dots, p)$ , and  $-1$  if it takes an odd number.

For  $p = 2$ , the columns have two orderings. The ordering  $(1, 2)$  keeps each column in place, picks the entries  $a_{11}$  and  $a_{22}$ , and has sign  $+1$ . The ordering  $(2, 1)$  swaps the two columns, picks the entries  $a_{12}$  and  $a_{21}$ , and has sign  $-1$ . The sum  $a_{11}a_{22} - a_{12}a_{21}$  is the value  $ad - bc$  from Example 0.33.

See also <https://en.wikipedia.org/wiki/Determinant>.

**Theorem 0.26** (Determinant of a diagonal matrix). *The determinant (Definition 0.51) of a  $p \times p$  diagonal matrix (Definition 0.34) is the product of its diagonal entries:*

$$\det \left( \begin{bmatrix} d_1 & \cdots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \cdots & d_p \end{bmatrix} \right) = \prod_{i=1}^p d_i$$

### **i** Proof

*Proof.* By induction on  $p$ . For  $p = 1$ , the matrix is  $[d_1]$ , and its determinant is  $d_1$  by definition.

For  $p \geq 2$ , suppose the result holds for  $(p-1) \times (p-1)$  diagonal matrices, and let  $\mathbf{D}$  be  $p \times p$  and diagonal. Row 1 of  $\mathbf{D}$  is  $(d_1, 0, \dots, 0)$ , so only the  $j = 1$  term of the definition can be nonzero, and deleting row 1 and column 1 of  $\mathbf{D}$  leaves the diagonal matrix with diagonal entries  $d_2, \dots, d_p$ :

$$\begin{aligned} \det(\mathbf{D}) &= (-1)^{1+1} d_1 \det(\mathbf{D}_{(11)}) + \sum_{j=2}^p (-1)^{1+j} \cdot 0 \cdot \det(\mathbf{D}_{(1j)}) && \text{(definition)} \\ &= d_1 \det(\mathbf{D}_{(11)}) && \text{(drop the zero terms)} \\ &= d_1 \prod_{i=2}^p d_i && \text{(induction hypothesis)} \\ &= \prod_{i=1}^p d_i && \text{(combine)} \end{aligned}$$

□

**Example 0.34** (Determinant of a scaled identity matrix). For a number  $c$ ,  $c\mathbf{I}_p$  is diagonal with every diagonal entry equal to  $c$ , so  $\det(c\mathbf{I}_p) = c^p$ . In particular,  $\det(\mathbf{I}_p) = 1$ .

**Theorem 0.27** (Determinant of a product). For  $p \times p$  matrices  $\mathbf{A}$  and  $\mathbf{B}$ ,

$$\det(\mathbf{AB}) = \det(\mathbf{A}) \det(\mathbf{B}).$$

**Example 0.35** (Checking the product rule). For  $\mathbf{A} = \begin{bmatrix} 2 & 1 \\ 0 & 1 \end{bmatrix}$  and  $\mathbf{B} = \begin{bmatrix} 1 & 0 \\ 3 & 1 \end{bmatrix}$ , Example 0.33 gives  $\det(\mathbf{A}) = 2 \cdot 1 - 1 \cdot 0 = 2$  and  $\det(\mathbf{B}) = 1 \cdot 1 - 0 \cdot 3 = 1$ . The product is

$$\mathbf{AB} = \begin{bmatrix} 2 \cdot 1 + 1 \cdot 3 & 2 \cdot 0 + 1 \cdot 1 \\ 0 \cdot 1 + 1 \cdot 3 & 0 \cdot 0 + 1 \cdot 1 \end{bmatrix} = \begin{bmatrix} 5 & 1 \\ 3 & 1 \end{bmatrix},$$

with  $\det(\mathbf{AB}) = 5 \cdot 1 - 1 \cdot 3 = 2 = \det(\mathbf{A}) \det(\mathbf{B})$ .

*Remark 0.34* (An example is not a proof). Example 0.35 checks Theorem 0.27 for one pair of  $2 \times 2$  matrices, which shows the theorem holds there but does not prove it for every pair. The proof needs properties of the determinant that these notes don't develop; see Banerjee and Roy (2014).

See also <https://en.wikipedia.org/wiki/Determinant>.

**Theorem 0.28** (The determinant of a symmetric matrix is the product of its eigenvalues). *Let  $\mathbf{A}$  be a  $p \times p$  symmetric matrix with real entries, with eigendecomposition  $\mathbf{A} = \mathbf{Q}\mathbf{Q}^\top$  (Definition 0.47) and eigenvalues  $\lambda_1, \dots, \lambda_p$  on the diagonal of  $\mathbf{\Lambda}$ . Then*

$$\det(\mathbf{A}) = \prod_{i=1}^p \lambda_i.$$

### Proof

*Proof.* First,  $\det(\mathbf{Q}^\top) \det(\mathbf{Q}) = 1$ :

$$\begin{aligned} \det(\mathbf{Q}^\top) \det(\mathbf{Q}) &= \det(\mathbf{Q}^\top \mathbf{Q}) && \text{(determinant of a product)} \\ &= \det(\mathbf{I}_p) && \text{(\mathbf{Q} is orthogonal)} \\ &= 1 && \text{(determinant of the identity)} \end{aligned}$$

Then:

$$\begin{aligned} \det(\mathbf{A}) &= \det(\mathbf{Q}\mathbf{Q}^\top) && \text{(substitute the eigendecomposition)} \\ &= \det(\mathbf{Q}) \det(\mathbf{\Lambda}) \det(\mathbf{Q}^\top) && \text{(determinant of a product, twice)} \\ &= \det(\mathbf{\Lambda}) \cdot \det(\mathbf{Q}^\top) \det(\mathbf{Q}) && \text{(reorder the three numbers)} \\ &= \det(\mathbf{\Lambda}) && \text{(the first display)} \\ &= \prod_{i=1}^p \lambda_i && \text{(determinant of a diagonal matrix)} \end{aligned}$$

The product steps are Theorem 0.27, the orthogonality step is Definition 0.40, and the identity and diagonal steps are Theorem 0.26 and Example 0.34.  $\square$

**Corollary 0.2** (A positive definite matrix has a positive determinant). *If  $\mathbf{A}$  is positive definite (Definition 0.50), then  $\det(\mathbf{A}) > 0$ .*

**i** Proof

*Proof.* By Theorem 0.24, every eigenvalue of  $\mathbf{A}$  is positive, so their product, which is  $\det(\mathbf{A})$  by Theorem 0.28, is positive.  $\square$

**Example 0.36** (Two ways to the same determinant).  $\mathbf{A} = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$  has eigenvalues 3 and 1 (Example 0.24), so Theorem 0.28 gives  $\det(\mathbf{A}) = 3 \cdot 1 = 3$ , matching  $2 \cdot 2 - 1 \cdot 1 = 3$  from Example 0.33.

## Design Matrix

**Definition 0.52** (Design matrix). In a regression model with  $n$  observations and  $p$  predictors, the **design matrix** (or *model matrix*)  $\mathbf{X}$  is the  $n \times p$  matrix whose  $i$ -th row is the covariate vector  $\tilde{x}_i^\top$  for observation  $i$ :

$$\mathbf{X} = \begin{bmatrix} \tilde{x}_1^\top \\ \tilde{x}_2^\top \\ \vdots \\ \tilde{x}_n^\top \end{bmatrix} = \begin{bmatrix} x_{11} & x_{12} & \cdots & x_{1p} \\ x_{21} & x_{22} & \cdots & x_{2p} \\ \vdots & \vdots & \ddots & \vdots \\ x_{n1} & x_{n2} & \cdots & x_{np} \end{bmatrix}$$

*Remark 0.35* (Products with the design matrix). The product  $\mathbf{X}\tilde{\beta}$  collects the values  $\tilde{x}_i^\top \tilde{\beta}$  for all  $n$  observations into a single  $n \times 1$  vector:

$$\mathbf{X}\tilde{\beta} = \begin{bmatrix} \tilde{x}_1^\top \tilde{\beta} \\ \vdots \\ \tilde{x}_n^\top \tilde{\beta} \end{bmatrix}$$

For example, take the rank-2 matrix from Example 0.9 as  $\mathbf{X}$ , with  $n = 3$  observations and  $p = 2$  columns: an intercept column and one covariate. With  $\tilde{\beta} = (1, 2)$ ,

$$\mathbf{X}\tilde{\beta} = \begin{bmatrix} 1 & 1 \\ 1 & 2 \\ 1 & 3 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \end{bmatrix} = \begin{bmatrix} 3 \\ 5 \\ 7 \end{bmatrix}.$$

The matrix  $\mathbf{X}^\top \mathbf{X}$  is  $p \times p$  and symmetric, since  $(\mathbf{X}^\top \mathbf{X})^\top = \mathbf{X}^\top (\mathbf{X}^\top)^\top = \mathbf{X}^\top \mathbf{X}$  (Theorem 0.10). When  $\mathbf{X}^\top \mathbf{X}$  is invertible, it appears in the OLS estimator  $\hat{\tilde{\beta}} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \tilde{y}$ .

**Theorem 0.29** ( $\mathbf{X}^\top \mathbf{X}$  is invertible when  $\mathbf{X}$  has full column rank). *If  $\mathbf{X}$  is an  $n \times p$  matrix with  $\text{rank}(\mathbf{X}) = p$  (Definition 0.26), then the  $p \times p$  matrix  $\mathbf{X}^\top \mathbf{X}$  is invertible.*

### **i** Proof

*Proof.* Let  $\tilde{c}$  be a vector of length  $p$  with  $\mathbf{X}^\top \mathbf{X} \tilde{c} = \tilde{0}$ . Then

$$\begin{aligned} 0 &= \tilde{c}^\top \mathbf{X}^\top \mathbf{X} \tilde{c} && \text{(multiply } \mathbf{X}^\top \mathbf{X} \tilde{c} = \tilde{0} \text{ on the left by } \tilde{c}^\top) \\ &= (\mathbf{X} \tilde{c})^\top (\mathbf{X} \tilde{c}) && \text{(transpose of a product)} \\ &= \|\mathbf{X} \tilde{c}\|^2 && \text{(definition of the Euclidean norm)} \end{aligned}$$

so  $\mathbf{X} \tilde{c} = \tilde{0}$ . Because  $\mathbf{X} \tilde{c} = c_1(\text{column 1}) + \dots + c_p(\text{column } p)$  and the columns of  $\mathbf{X}$  are linearly independent,  $\tilde{c} = \tilde{0}$ . So the only solution of  $\mathbf{X}^\top \mathbf{X} \tilde{c} = \tilde{0}$  is  $\tilde{c} = \tilde{0}$ , which says that the columns of the square matrix  $\mathbf{X}^\top \mathbf{X}$  are linearly independent, and a square matrix with linearly independent columns is invertible (Banerjee and Roy 2014, Corollary 5.7, p. 143).  $\square$

**Example 0.37** (Inverting  $\mathbf{X}^\top \mathbf{X}$ ). For the rank-2 matrix  $\mathbf{X} = \begin{bmatrix} 1 & 1 \\ 1 & 2 \\ 1 & 3 \end{bmatrix}$  from Example 0.9:

$$\mathbf{X}^\top \mathbf{X} = \begin{bmatrix} 3 & 6 \\ 6 & 14 \end{bmatrix}, \quad (\mathbf{X}^\top \mathbf{X})^{-1} = \frac{1}{6} \begin{bmatrix} 14 & -6 \\ -6 & 3 \end{bmatrix},$$

and multiplying the two out gives  $\mathbf{I}_2$ .

**Definition 0.53** (Hat matrix). For an  $n \times p$  design matrix  $\mathbf{X}$  (Definition 0.52) with  $\text{rank}(\mathbf{X}) = p$ , the **hat matrix** is the  $n \times n$  matrix

$$\underset{n \times n}{\mathbf{H}} \stackrel{\text{def}}{=} \underset{n \times p}{\mathbf{X}} \underbrace{(\mathbf{X}^\top \mathbf{X})^{-1}}_{p \times p} \underbrace{\mathbf{X}^\top}_{p \times n}$$

The inverse exists by Theorem 0.29.

**Example 0.38** (The hat matrix of an intercept-only model). With  $n = 2$  observations and only an intercept,  $\mathbf{X} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$  ( $2 \times 1$ , rank 1), so  $\mathbf{X}^\top \mathbf{X} = 2$  and

$$\mathbf{H} = \begin{bmatrix} 1 \\ 1 \end{bmatrix} \cdot \frac{1}{2} \cdot \begin{bmatrix} 1 & 1 \end{bmatrix} = \begin{bmatrix} 0.5 & 0.5 \\ 0.5 & 0.5 \end{bmatrix}.$$

Then  $\mathbf{H} \tilde{y} = (\bar{y}, \bar{y})$ , where  $\bar{y} = (y_1 + y_2)/2$ : the fitted values of an intercept-only model are

the sample mean.

**Theorem 0.30** (Hat matrix is a projection matrix). *If  $\mathbf{X}$  is an  $n \times p$  design matrix with  $\text{rank}(\mathbf{X}) = p$ , then the hat matrix  $\mathbf{H}$  (Definition 0.53) is an orthogonal projection matrix (Definition 0.38).*

### i Proof

*Proof.* We verify symmetry and idempotency. Both use that  $\mathbf{X}^\top \mathbf{X}$  is symmetric,  $(\mathbf{X}^\top \mathbf{X})^\top = \mathbf{X}^\top (\mathbf{X}^\top)^\top = \mathbf{X}^\top \mathbf{X}$  (Theorem 0.10), so its inverse is symmetric too (Corollary 0.1).

**Symmetry:**

$$\begin{aligned}
 \mathbf{H}^\top &= (\mathbf{X}(\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top)^\top && \text{(definition of } \mathbf{H} \text{)} \\
 &= (\mathbf{X}^\top)^\top \cdot ((\mathbf{X}^\top \mathbf{X})^{-1})^\top \cdot \mathbf{X}^\top && \text{(transpose of a product, twice)} \\
 &= \mathbf{X} \cdot ((\mathbf{X}^\top \mathbf{X})^{-1})^\top \cdot \mathbf{X}^\top && \text{(transposing twice changes nothing)} \\
 &= \mathbf{X} \cdot (\mathbf{X}^\top \mathbf{X})^{-1} \cdot \mathbf{X}^\top && \text{(the inverse of a symmetric matrix is symmetric)} \\
 &= \mathbf{H} && \text{(definition of } \mathbf{H} \text{)}
 \end{aligned}$$

**Idempotency:**

$$\begin{aligned}
 \mathbf{H}^2 &= \mathbf{X}(\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \cdot \mathbf{X}(\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top && \text{(definition of } \mathbf{H} \text{)} \\
 &= \mathbf{X}(\mathbf{X}^\top \mathbf{X})^{-1} (\mathbf{X}^\top \mathbf{X}) (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top && \text{(regroup; matrix multiplication is associative)} \\
 &= \mathbf{X} \mathbf{I}_p (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top && \text{(definition of the inverse)} \\
 &= \mathbf{X}(\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top && \text{(identity matrix)} \\
 &= \mathbf{H} && \text{(definition of } \mathbf{H} \text{)}
 \end{aligned}$$

□

**Example 0.39** (The intercept-only hat matrix is a projection). For  $\mathbf{H} = \begin{bmatrix} 0.5 & 0.5 \\ 0.5 & 0.5 \end{bmatrix}$  from Example 0.38,  $\mathbf{H}^\top = \mathbf{H}$ , and

$$\mathbf{H}^2 = \begin{bmatrix} 0.5 \cdot 0.5 + 0.5 \cdot 0.5 & 0.5 \cdot 0.5 + 0.5 \cdot 0.5 \\ 0.5 \cdot 0.5 + 0.5 \cdot 0.5 & 0.5 \cdot 0.5 + 0.5 \cdot 0.5 \end{bmatrix} = \begin{bmatrix} 0.5 & 0.5 \\ 0.5 & 0.5 \end{bmatrix} = \mathbf{H},$$

as Theorem 0.30 says.

*Remark 0.36* (Why it is called the hat matrix). The hat matrix gives the fitted values in linear regression:  $\hat{\tilde{y}} = \mathbf{X}\hat{\tilde{\beta}} = \mathbf{X}(\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \tilde{y} = \mathbf{H}\tilde{y}$ . Multiplying by  $\mathbf{H}$  “puts a hat” on  $\tilde{y}$ , which is where the name comes from. In Example 0.38,  $\mathbf{H}$  puts a hat on  $\tilde{y}$  by replacing each  $y_i$  with the sample mean  $\bar{y}$ .

## Further reading

- Strang (2023) is a widely used first course in linear algebra. It covers vectors, elimination, vector spaces, orthogonality, determinants, eigenvalues, and the singular value decomposition.
- Axler (2024) is a proof-based treatment centered on vector spaces and linear maps. A free electronic version is available from the publisher.
- Fieller (2016)
- Banerjee and Roy (2014)
- Searle and Khuri (2017)

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